SERVICE REQUEST AND WORKFLOW AUTOMATION SYSTEM

A Project Report Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project work entitled "**SERVICE REQUEST AND WORKFLOW AUTOMATION SYSTEM**" is being submitted for the partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering, at BVC Engineering College, Odalarevu**, is a bonafide work done by **D.DURGA PRASAD(23225A0505),P.KASI(23225A0510),Y.V.D.SAI(23225A0512),K.SAIRAM(23225A0506),S.S.R.CH.PHANI KUMAR(23225A0511)** under my guidance during the academic year 2022-2026 and it has been found suitable for acceptance according to the requirement of the University.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree.

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ABSTRACT

In modern organizations, managing service requests efficiently is a critical challenge due to increasing workloads, multiple approval levels, and the dependency on manual or semi-manual processes such as emails, spreadsheets, and paper-based forms. These traditional approaches often lead to delays, lack of transparency, poor tracking, and increased chances of human error. To address these challenges, the Service Request & Workflow Automation System has been designed and developed as a comprehensive web-based solution that automates the complete lifecycle of service requests within an organization.

The proposed system provides a centralized digital platform where users can raise service requests related to various organizational services through a simple and user-friendly interface. Each request is automatically routed through a predefined workflow that includes request validation, approval, task assignment, execution, and closure. The system ensures that all stakeholders involved in the workflow are notified at every stage through automated alerts, thereby improving communication and reducing response time.

Administrators are provided with advanced features to manage users, define service categories, configure workflow rules, assign roles, and monitor request progress in real time. The system also maintains detailed logs and historical records of all requests, enabling effective auditing and performance evaluation. Analytical dashboards and reports help management identify bottlenecks, measure service efficiency, and make data-driven decisions.

The application is developed using a Java Full Stack architecture, incorporating a robust backend built with Spring Boot and RESTful APIs, a secure and scalable MySQL database, and a responsive frontend interface using modern web technologies. The modular design of the system allows easy scalability, customization, and integration with other enterprise applications in the future.

LIST OF CONTENTS……

CONTENTS **PAGE NO**

CHAPTER-1 INTRODUCTION **1**

CHAPTER - 2 LITERATURE SURVEY **2**

CHAPTER - 3 SYSTEM ANALYSIS **3**

3.1 Existing System 3

3.1.1 Drawbacks 4

3.2 Proposed System 4

3.2.1 Advantages 5

CHAPTER – 4 REQUIREMENT ANALYSIS **6**

4.1 Hardware Requirements 6

4.2 Software Requirements 6

CHAPTER - 5 SOFTWARE DESIGN **7**

5.1 System Architecture 7

5.2 UML Diagrams 7

5.2.1 Class Diagram 7

5.2.2 Use Case Diagram 8

5.2.3 Sequence Diagram 9

CHAPTER - 6 SYSTEM ENVIRONMENT **10**

6.1 What is JAVA 11

6.2 Advantages of Java over Other Languages 12

6.3 Disadvantages of Java 13

6.4 History of Java 14

6.5 What is Java Spring Boot 14

6.6 Categories of Java Spring Boot 15

6.7 Challenges in Java Spring Boot 16

6.8 Applications of Java Spring Boot 17

6.9 JAVA Development Steps 22

CHAPTER – 7 MODULES USED IN PROJECT **24**

7.1 Install Java Step-by-Step in Windows or MAC 25

7.2 How to Install Java on Windows and Mac 25

CHAPTER – 8 TESTING **35**

8.1 System Test 35

8.2 Types Of Tests 35
8.2.1 Unit testing 35
8.2.2 Integrated Testing 35
8.2.3 Functional test 36
8.2.4 White Box Testing 36
8.2.5 Black Box Testing 36
8.2.6 Unit Testing 36
CHAPTER - 9 IMPLEMENTATION **38**
9.1 Methodology 38
9.2 Source Code 40
CHAPTER – 10 RESULTS **45**
CHAPTER – 11 CONCLUSION **54**
CHAPTER – 12 REFERENCES **55**

LIST OF FIGURES

FIGURES **PAGE NO**

Fig 5.1 System Architecture....7
Fig 5.2 Class Diagram....8
Fig 5.3 Use Case Diagram....8
Fig 5.4 Sequence Diagram....9
Fig 7.1 Open Java website to download the Java latest version....25
Fig 7.2 Click on Download Java....26
Fig 7.3 Java Development Kit (JDK) Setup File....27
Fig 7.4 JDK Installation Path configuration....27
Fig 7.5 JDK Setup Process in Progress....28
Fig 7.6 Successful Installation of Java Development Kit (JDK)....28
Fig 7.7 Java Environment Path Configuration....29
Fig 7.8 Accessing Environment Variables in System Properties....29
Fig 7.9 Environment Variables Configuration for Java....30
Fig 7.10 Configuration of System PATH Environment Variable....30
Fig 7.11 Verifying Installed Java Version via Command Line....31
Fig 7.12 Basic Java Program to Print “Hello World”....31
Fig 7.13 Command-Line Verification of Java Installation and Drive Navigation....32
Fig 7.14 Directory Listing in Windows Command Prompt....32
Fig 7.15 Compilation and Execution of a Java Program via Command Line....33
Fig 10.1 User Login Screen....45
Fig10.2 User Registration Screen....46
Fig 10.3 Spring Boot Server Startup....46
Fig 10.4 User Dashboard....47
Fig 10.5 Create New Request – Category Selection....47
Fig 10.6 My Service Requests....48
Fig 10.7 Admin Dashboard....48
Fig 10.8 User Management....49
Fig 10.9 Service Catalog Management....50
Fig 10.10 Category Management....50
Fig 10.11 Reports & Analytics....51
Fig 10.12 System Settings – Email Configuration....52
Fig 10.13 My Service Requests – Detailed View....53

CHAPTER-1

INTRODUCTION

In the present digital era, organizations increasingly depend on information systems to manage daily operations efficiently. One of the most important operational activities in any organization is the handling of service requests such as IT support issues, maintenance tasks, administrative approvals, and technical assistance. When these requests are managed manually or through unstructured methods, it often results in delays, lack of accountability, and inefficient utilization of resources.

Traditionally, service requests were handled using paper forms, emails, or verbal communication. These methods do not provide a centralized mechanism to track the status of requests or monitor their progress. As organizations grow in size, the volume of service requests increases significantly, making manual handling difficult and error-prone. Delays in response, miscommunication between departments, and the absence of proper documentation are common problems associated with traditional systems.

To address these issues, organizations began adopting basic computerized systems to record service requests. Although these systems reduced paperwork and improved data storage, they still lacked structured workflows and automation. Requests often required manual approvals, and task assignments depended on individual decision-making. This resulted in inconsistent processing, increased workload on administrators, and reduced transparency for users.

The Service Request Workflow Automation System is proposed to overcome the limitations of existing systems by providing a centralized, automated, and role-based platform for managing service requests. The system allows users to submit service requests through an online interface, where each request is recorded, tracked, and processed through predefined workflow stages. These stages include request submission, validation, approval, processing, and closure.

The system is designed with role-based access control, ensuring that users, administrators, and service personnel have clearly defined responsibilities. Automated workflows reduce manual intervention and ensure that requests are routed to the appropriate authorities without delay. The system also provides real-time status updates, enabling users to monitor the progress of their requests and improving overall transparency.

Developed using modern web technologies, the system employs a robust backend framework and a reliable database to ensure scalability, security, and data integrity. The application can be used across various domains such as educational institutions, corporate offices, IT service departments, and public sector organizations. By automating service request handling, the system improves operational efficiency, reduces response time, and enhances user satisfaction.

CHAPTER-2

LITERATURE SURVEY

A literature survey is an essential phase in software development that involves studying existing systems, research papers, and technologies related to the proposed project. The main objective of this survey is to understand current approaches used in service request management, identify their limitations, and explore techniques that can be improved through automation and modern web technologies.

In many organizations, service request handling has traditionally been performed using manual methods such as paper-based forms, emails, or verbal communication. These methods lack standard procedures and centralized control. According to various studies, manual systems lead to delays in request processing, difficulty in tracking request status, duplication of work, and increased chances of human error. As the number of users and service requests increases, manual handling becomes inefficient and unmanageable.

To overcome these challenges, basic computerized ticket management systems were introduced. These systems allow users to submit service requests electronically and enable administrators to assign tasks to service personnel. While these systems improved data storage and reduced paperwork, they still required significant manual intervention. Most early systems did not support automated approvals, rule-based task routing, or real-time monitoring, which limited their effectiveness in complex organizational environments.

Recent literature highlights the growing importance of workflow automation in service management systems. Workflow automation involves defining a sequence of actions that a service request must follow, such as submission, validation, approval, execution, and closure. Research indicates that automated workflows improve operational efficiency by ensuring consistency, reducing response time, and minimizing dependency on individuals. Role-based access control is also emphasized as a key feature to enhance security and accountability by restricting system access based on user roles.

With advancements in web application development, frameworks such as Spring Boot enable the creation of scalable, secure, and modular backend systems. Studies show that database-driven applications using relational databases like MySQL provide reliable storage, data integrity, and efficient query handling. Integration of frontend technologies with backend services allows real-time status updates and improved user interaction.

Despite the availability of commercial workflow automation tools, many existing solutions are expensive, complex, and difficult to customize. Small and medium-scale organizations often require a simple, flexible, and cost-effective system that can be adapted to their specific workflows

CHAPTER - 3
SYSTEM ANALYSIS

3.1 Existing System

In many organizations, service request management is primarily handled using traditional manual or semi-automated methods. Users typically raise service requests through emails, phone calls, paper forms, or informal communication channels. Once a request is submitted, it is forwarded manually to the concerned department or authority for further action. This approach lacks a structured workflow and depends heavily on human intervention at every stage of request processing.

In some cases, basic computerized systems or simple ticketing tools are used to record service requests. Although these systems help in storing request details electronically, they do not provide complete workflow automation. The process of approval, task assignment, escalation, and closure is often performed manually by administrators. As a result, there is no standardized method for handling different types of service requests.

The existing system does not provide real-time tracking of request status for users. Once a request is submitted, users are required to follow up repeatedly to know the progress of their request. This lack of transparency leads to user dissatisfaction and inefficient communication between departments. Moreover, the absence of automated notifications further increases delays in service delivery.

Another major limitation of the existing system is the lack of role-based access control. Since responsibilities are not clearly defined, unauthorized access and data inconsistency may occur. Reporting and analysis of service requests are also difficult because data is scattered across multiple platforms or maintained manually.

1. Traditional Manual Request Handling Techniques

In the existing system, service requests are handled using traditional manual techniques such as paper forms, emails, phone calls, or direct communication with staff members. Users submit their requests without a standardized digital platform, and the requests are processed manually by administrators or concerned departments. This approach lacks automation and relies completely on human intervention for forwarding, approval, and execution of requests.

2. Email-Based or Semi-Automated Request Systems

Some organizations use email-based systems or simple digital forms to collect service requests. In this approach, users send requests through email or basic online forms, and administrators manually review and assign tasks. Although this method reduces paperwork, it still requires manual monitoring and decision-making at each stage.

3. Lack of Workflow Automation

The existing system does not implement workflow automation techniques. Each request is processed independently without predefined rules or stages. Approvals, assignments, and closures depend on individual judgment rather than system-driven logic. This leads to delays, unequal workload distribution, and lack of accountability.

4. Absence of Role-Based Access Control

In the existing system, roles and responsibilities are not clearly defined. Users, administrators, and service staff may have overlapping access, which can lead to unauthorized actions or data inconsistency. Security and accountability are compromised due to the lack of proper access control mechanisms.

5. Limited Reporting and Monitoring

Existing systems do not provide proper reporting and analytics features. Since data is maintained manually or across multiple platforms, generating reports on request status, response time, or workload is time-consuming and inaccurate.

3.1.1 Drawbacks

The existing service request handling system suffers from several limitations that affect overall efficiency and reliability. One of the major drawbacks is the heavy dependence on manual processes. Since requests are handled through paper forms, emails, or verbal communication, the system is time-consuming and prone to human errors. Requests may be misplaced, delayed, or even ignored due to improper tracking mechanisms.

Another significant drawback is the lack of workflow automation. In the absence of predefined workflows, service requests do not follow a structured sequence of steps. Approvals, task assignments, and escalations are performed manually, which leads to inconsistency and delays in request resolution. This also increases the workload on administrators and reduces operational efficiency.

The existing system also lacks transparency. Users are unable to track the real-time status of their service requests and must depend on follow-ups to receive updates. This results in poor user experience and dissatisfaction. Additionally, the absence of automated notifications further delays communication between users and service staff.

3.2 Proposed System

The proposed Service Request Workflow Automation System is designed to overcome the limitations of the existing manual and semi-automated systems by introducing a centralized, automated, and role-based platform for managing service requests. The system provides a structured approach to handle service requests efficiently from submission to closure using predefined workflows.

In the proposed system, users can submit service requests through a web-based interface by providing necessary details such as request type, description, and priority. Once a request is submitted, it is automatically recorded in the system and assigned a

unique request ID. The request is then routed through predefined workflow stages such as validation, approval, processing, and completion without manual intervention.

The system incorporates workflow automation, where business rules control the movement of requests between different stages. This ensures consistency in request handling and reduces dependency on individual decision-making. Administrators have the ability to configure workflows, assign tasks to service staff, approve or reject requests, and monitor system performance through an admin dashboard.

3.2.1 Advantages:

The proposed Service Request Workflow Automation System offers several advantages over the existing manual and semi-automated systems. One of the primary advantages is the automation of service request handling. By using predefined workflows, the system reduces manual intervention and ensures that each request follows a structured sequence of steps from submission to completion. This significantly decreases processing time and improves overall efficiency.

Another major advantage is improved transparency and real-time tracking. Users can monitor the status of their service requests at any stage, eliminating the need for repeated follow-ups. Automated notifications keep users, administrators, and service staff informed about important updates such as approvals, task assignments, and completion of requests, which enhances communication and user satisfaction.

The system also provides enhanced security through role-based access control. Each user is assigned specific permissions based on their role, ensuring that sensitive data and critical operations are accessible only to authorized individuals. This improves accountability and reduces the risk of unauthorized access or data misuse.

CHAPTER – 4

REQUIREMENT ANALYSIS

4.1 Hardware Requirements

- **Processor:** Intel Core i3 or above
- **RAM:** Minimum 4 GB
- **Hard Disk:** Minimum 250 GB
- **Internet Connection:** Required
- **Input Devices:** Keyboard, Mouse
- **Output Device:** Monitor

4.2 Software Requirements

- **Operating System:** Windows 10 / Linux
- **Frontend Technologies:** HTML, CSS, Bootstrap, JavaScript / React
- **Backend Technology:** Java, Spring Boot, REST APIs
- **Database:** MySQL
- **Web Server:** Apache Tomcat
- **Development Tools:** Eclipse / IntelliJ IDEA, Postman, VS code
- **Browser:** Google Chrome / Mozilla Firefox

CHAPTER - 5

SOFTWARE DESIGN

5.1 System Architecture

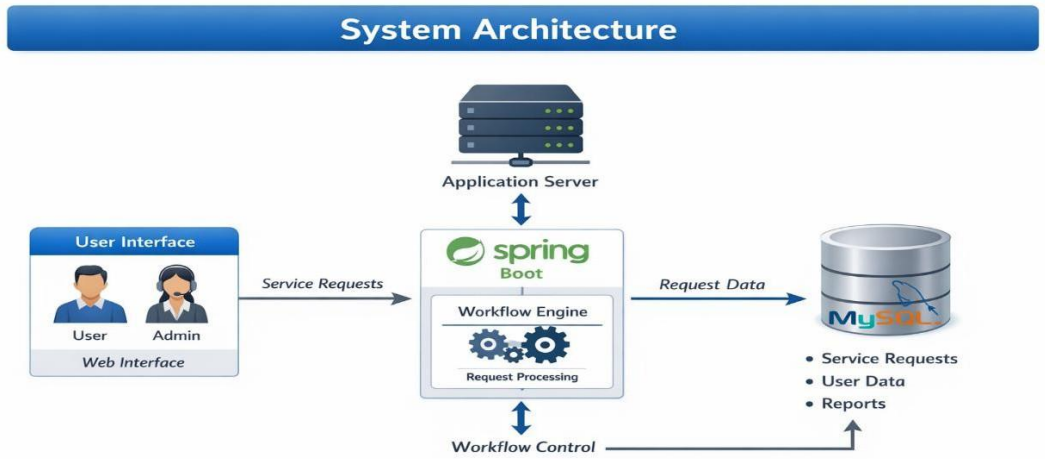


Fig 5.1 System Architecture

5.2 UML Diagrams

5.2.1 Class Diagram

The class diagram represents the static structure of the Service Request Workflow Automation System. It shows the main classes, their attributes, methods, and the relationships between them. This diagram helps in understanding how data and functionality are organized within the system.

User Class

The User class represents individuals who interact with the system, such as employees or service requesters. It contains attributes like user ID, name, email, password, and role. The methods defined in this class allow users to log in, log out, raise service requests, and view the status of their requests.

Service Request Class

The ServiceRequest class is the core entity of the system. It stores details related to a service request, including request ID, type, description, priority, status, and creation date. This class provides methods to create new requests, update request status, and close completed requests. Each service request is associated with a user.

Workflow Class

The Workflow class manages the movement of service requests through different stages. It contains information about the current stage and the next stage of a request. Methods in this class control stage transitions and validate workflow rules. This ensures that every request follows a predefined and structured process.

Admin Class

The Admin class represents administrative users responsible for managing service requests. Admins can approve or reject requests, assign tasks to service staff, and monitor

workflow progress. This class plays a crucial role in enforcing rules and ensuring timely request processing.

Notification Class

The Notification class is used to inform users and administrators about changes in request status. It stores notification details such as message content and date. The send notification method ensures that stakeholders receive timely updates regarding approvals, assignments, or completion of requests.

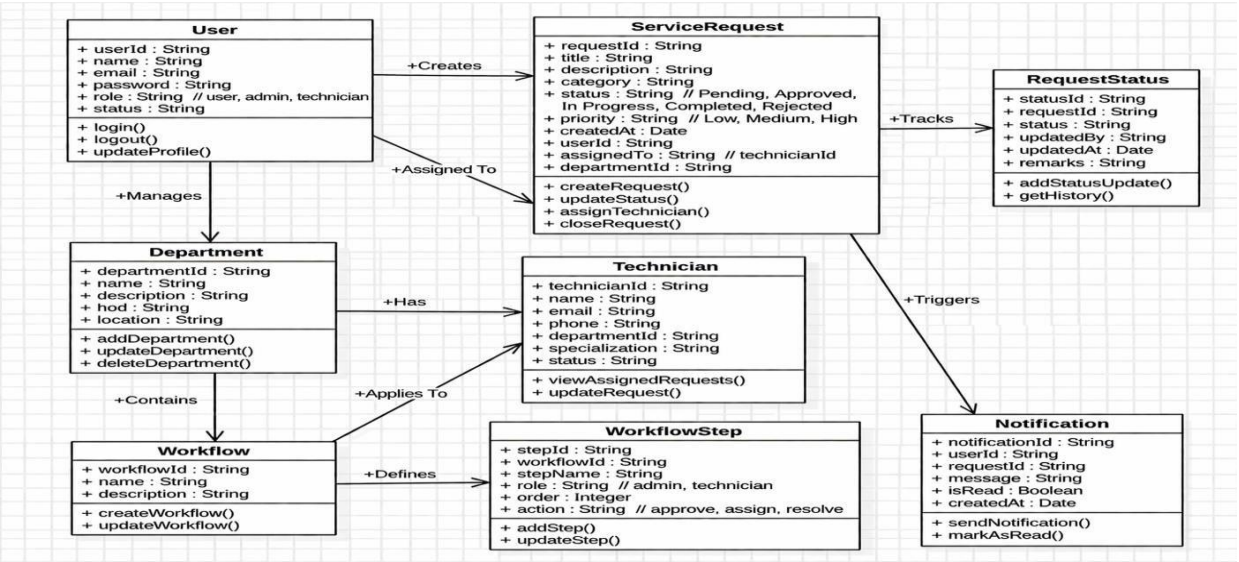


Fig 5.2 Class Diagram

5.2.2 Use Case Diagram

A use case diagram represents the functional behavior of the system by showing interactions between users (actors) and the system. It helps in understanding what the system does rather than how it is implemented. The use case diagram for the Service Request and Workflow Automation System illustrates the various operations performed by different actors involved in the system

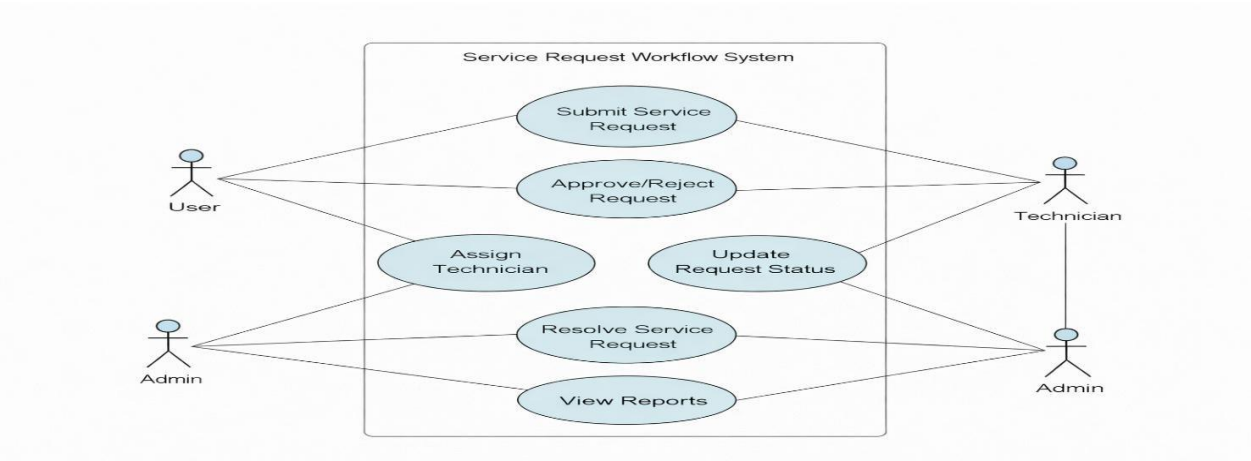


Fig 5.3 Use Case Diagram

5.2.3 Sequence Diagram

The sequence diagram illustrates the interaction between different system components over time in the Service Request and Workflow Automation System. It represents the chronological order of messages exchanged between the user, application system, administrator, service staff, and database during the processing of a service request. This diagram helps in understanding how a service request flows through the system from initiation to completion.

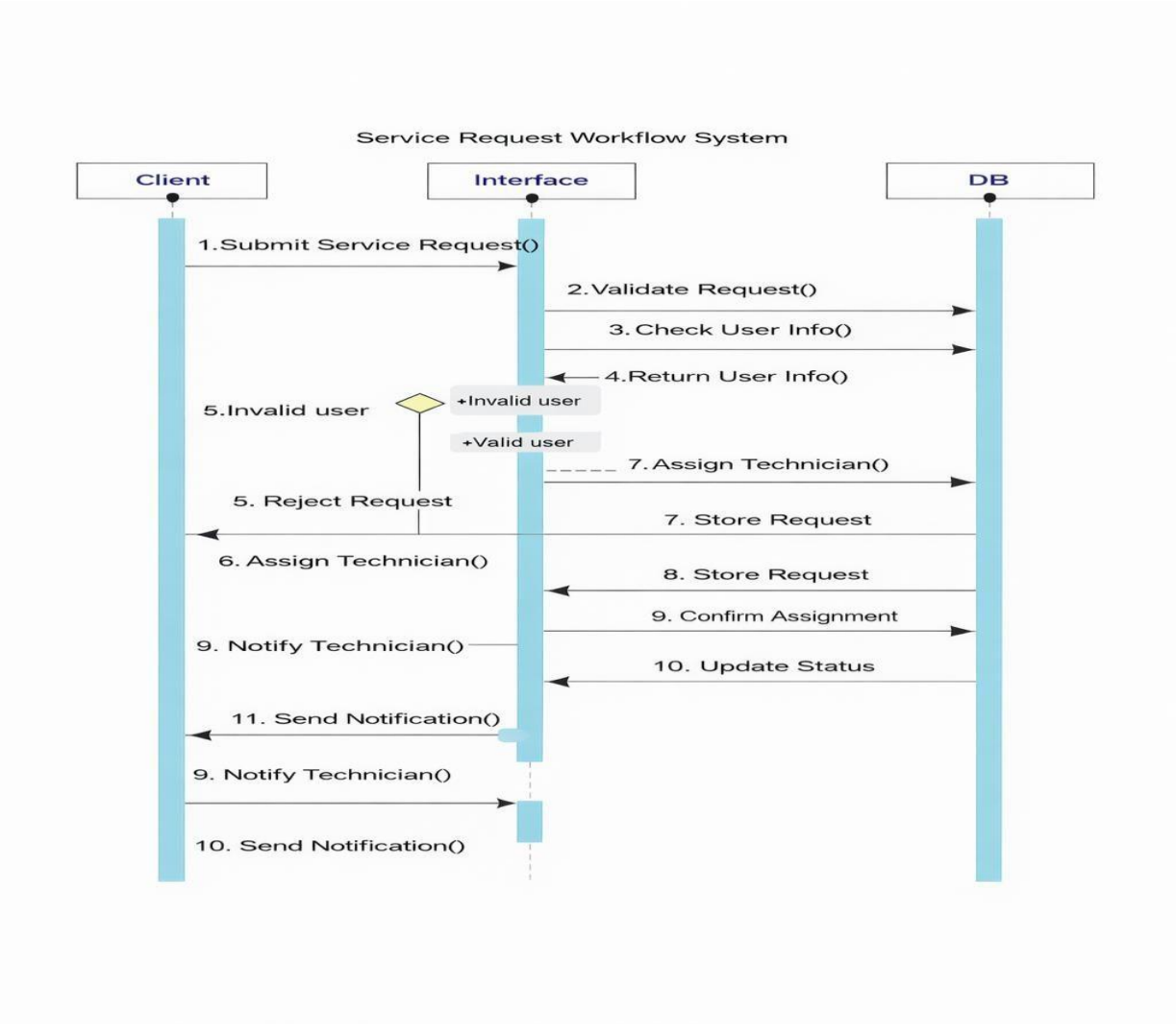


Fig 5.4 Sequence Diagram

CHAPTER - 6

SYSTEM ENVIRONMENT

6.1 What is Java

- Java was developed by Sun Microsystems in 1995 and later acquired by Oracle Corporation.

- Java is a high-level, object-oriented programming language designed to be simple, secure, and portable.

- One of Java's core principles is "Write Once, Run Anywhere (WORA)", meaning Java programs can run on any platform that has a Java Virtual Machine (JVM).

- Java programs are compiled into bytecode, which is executed by the JVM, not directly by the operating system.

- Java is platform-independent, unlike languages such as C and C++, which are platform-dependent.

- It supports automatic memory management through Garbage Collection, reducing memory leaks and programmer effort.

1. **Platform Independence**:
   Java follows the principle "Write Once, Run Anywhere". Java code is compiled into bytecode, which runs on the Java Virtual Machine (JVM). As long as a system has a JVM, the same Java program can run on Windows, Linux, or macOS- without Modification

2. **Strong Presence in Enterprise & Large Systems**

Java is the industry standard for:
- Banking systems
- Insurance platforms
- E-commerce backends
- Government and defense applications

Enterprises prefer Java because it is:
- Stable over decades
- Backward compatible
- Designed for long-running applications

3. **Robust Object-Oriented Design**

Java strictly follows Object-Oriented Programming (OOP):
- Encapsulation
- Inheritance
- Polymorphism

•Abstraction

This makes Java applications:

•Easier to maintain
•Easier to scale
•Suitable for large team development

4. Excellent Memory Management

Java provides automatic garbage collection, which:

•Frees unused memory automatically
•Prevents many memory leaks
•Reduces crashes caused by manual memory handling

5. Built-in Security

Java was designed with security in mind:

•Bytecode verification
•JVM sandboxing
•Strong access control
•No direct pointer manipulation

6. Massive Ecosystem & Framework Support

Java has one of the largest ecosystems:

- Spring / Spring Boot
- Hibernate / JPA
- Maven / Gradle

7. Multithreading & Performance

Java has native multithreading support, which allows:

- High-performance concurrent processing
- Efficient CPU utilization
- Scalable server applications

8. Long-Term Support & Backward Compatibility

- Java prioritizes stability over breaking changes.
- Old Java code still runs on new JVM versions
- Enterprises can upgrade safely

9. Strong Job Market & Community

Java has:

- Millions of developers worldwide
- Huge open-source support
- Strong demand in jobs and interviews

6.2 Advantages of Java over Other Languages

1. Platform Independence
Java follows the principle "Write Once, Run Anywhere".
Programs are compiled into bytecode and executed by the Java Virtual Machine (JVM), allowing Java applications to run on any operating system without modification.

2. Strong Object-Oriented Programming
Java is a pure object-oriented language (except primitive types), supporting:
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction

3. Automatic Memory Management
Java provides Garbage Collection, which automatically removes unused objects from memory.
This reduces:
- Memory leaks
- Manual memory errors
- Program crashes

4. High Security
Java is designed with security as a core feature:
- No direct pointer access
- Bytecode verification
- JVM sandboxing
- Strong access control

5. Robust and Reliable
Java emphasizes compile-time checking, exception handling, and strong typing, which helps in:
- Early error detection
- Stable long-running applications
- Reduced runtime failures

6. Multithreading Support
Java has **built-in multithreading**, allowing multiple tasks to run concurrently.
This improves:
- Application performance
- CPU utilization
- Responsiveness in server-side applications

6.3 Disadvantages of Java

1. Slower Performance Compared to Low-Level Languages

Java programs run on the Java Virtual Machine (JVM), which adds an extra execution layer.

As a result, Java is generally slower than compiled languages like C and C++ for performance

2. High Memory Consumption

Java applications require:

- JVM
- Garbage Collector
- Runtime libraries

3. Verbose Code

Java requires more lines of code to perform simple operations.

This makes development:

- Slower for small applications
- Less concise compared to languages like Python or JavaScript

4. Not Ideal for Low-Level Programming

Java does not support:

- Direct memory access
- Pointer arithmetic

5. Slower Startup Time

Java applications often have a **slower startup time** due to:

- JVM initialization
- Class loading

6. Limited Use in AI and Data Science

Although Java is capable, it is **less preferred** for:

- Machine learning
- Data science
- Rapid AI experimentation

7. Garbage Collection Overhead

Automatic garbage collection can sometimes:

- Cause unpredictable pauses
- Reduce real-time performance

8. Dependency on JVM

Java programs cannot run directly on the operating system; they require a JVM.

If the JVM is unavailable or incompatible, the application cannot execute.

6.4 History of Java

Java was developed in the early 1990s by James Gosling and his team at Sun Microsystems. The project began in 1991 as part of the *Green Project*, whose goal was to design a programming language for consumer electronic devices such as set-top boxes and smart appliances. The original language was named Oak, inspired by an oak tree outside Gosling's office, but the name was later changed due to trademark issues.

The language was officially released to the public in 1995 under the name Java. Its key innovation was platform independence, achieved by compiling programs into bytecode that runs on the Java Virtual Machine (JVM). This led to the famous principle "Write Once, Run Anywhere", which quickly made Java popular as the World Wide Web expanded. Early adoption was driven by Java applets, which allowed interactive programs to run inside web browsers (later phased out due to security and technology changes).

In 2010, Sun Microsystems was acquired by Oracle Corporation, which took over stewardship of Java. Under Oracle, Java continued to evolve with regular releases, performance improvements, enhanced security, and long-term support (LTS) versions to meet enterprise needs.

Over time, Java expanded far beyond browsers and embedded devices. It became a dominant language for enterprise applications, banking systems, backend web services, cloud computing, and Android mobile development. Thanks to its stability, backward compatibility, strong ecosystem, and large developer community, Java remains one of the most widely used and trusted programming languages in the software industry today.

6.5 What is Java Spring Boot

Java Spring Boot is a modern framework used to develop standalone, production-ready Java applications with minimal configuration. It is built on top of the Spring Framework and is designed to simplify application development by reducing boilerplate code and configuration complexity. Spring Boot enables developers to create web applications, REST APIs, and enterprise systems efficiently while following industry best practices.

Spring Boot follows the concept of convention over configuration, which means it provides default configurations for common application requirements. This allows developers to focus more on business logic rather than spending time on XML files or manual setup. With features like auto-configuration, Spring Boot automatically configures components such as databases, security, and web servers based on the dependencies added to the project.

One of the key features of Spring Boot is its support for embedded servers such as Tomcat, Jetty, and Undertow. Because of this, applications can be run as standalone Java programs without the need to deploy them on an external server. This makes development,

testing, and deployment faster and easier, especially in modern DevOps and cloud environments.

Spring Boot provides excellent support for building RESTful web services. It integrates smoothly with Spring MVC and allows developers to create APIs using simple annotations. This makes it highly suitable for backend development in web applications, mobile applications, and microservices architectures. Spring Boot also supports JSON-based communication, which is widely used in modern applications.

Another important advantage of Spring Boot is its strong integration with databases and persistence technologies. It works seamlessly with JPA, Hibernate, and JDBC, enabling efficient data access and management. Spring Boot also supports transaction management and simplifies database connectivity, which is essential for enterprise applications.

6.6 Categories of Java Spring Boot

Java Spring Boot can be understood through several important categories, each focusing on a specific aspect of application development. These categories explain how Spring Boot supports building complete, scalable, and production-ready applications.

Spring Boot can first be categorized as a Web Application Framework. It is widely used to develop web applications and RESTful services. With built-in support for Spring MVC and embedded servers like Tomcat, Spring Boot allows developers to create web-based applications that can run independently without requiring an external application server. This category is especially useful for developing backend services for websites and mobile applications.

Another important category of Spring Boot is Microservices Development. Spring Boot is one of the most preferred frameworks for building microservices-based architectures. It allows applications to be broken into small, independent services that communicate with each other using REST APIs. Features such as lightweight configuration, fast startup, and easy integration with cloud platforms make Spring Boot ideal for microservices.

Spring Boot can also be categorized under Database and Persistence Support. It provides seamless integration with database technologies such as JPA, Hibernate, and JDBC. Spring Boot simplifies database configuration and enables developers to perform data operations efficiently. This category plays a crucial role in applications that require secure and structured data storage.

Another category is Security and Authentication. Spring Boot integrates with Spring Security to provide authentication, authorization, and role-based access control. It supports secure login mechanisms, password encryption, and protection against common security threats. This makes Spring Boot suitable for applications that require strong security features, such as enterprise and banking systems.

Need for Spring Boot

The need for Java Spring Boot arises from the challenges developers faced while building large and complex Java applications using traditional frameworks. Earlier Java development required extensive configuration, large XML files, and complex setup processes, which increased development time and reduced productivity. Spring Boot was introduced to simplify this process and make Java application development faster, easier, and more efficient.

One of the primary needs for Spring Boot is to reduce configuration complexity. Spring Boot follows the principle of convention over configuration, providing default settings for commonly used components. This eliminates the need for writing lengthy configuration files and allows developers to focus on implementing business logic rather than infrastructure setup.

Spring Boot is needed for rapid application development. With features like auto-configuration, starter dependencies, and embedded servers, applications can be developed and run quickly. Developers can create production-ready applications in less time compared to traditional Java frameworks, which is especially important in modern software development environments.

Another important need for Spring Boot is easy deployment. Since Spring Boot supports embedded servers such as Tomcat, applications can be run as standalone Java programs. This simplifies deployment and avoids dependency on external application servers, making Spring Boot suitable for cloud and DevOps-based environments.

6.7 Challenges in Java Spring Boot

- **Steep Learning Curve**
  Spring Boot is built on top of the Spring Framework, which has many concepts such as dependency injection, annotations, and inversion of control. Beginners may find it difficult to understand these concepts initially.

- **Hidden Auto-Configuration Complexity**
  Spring Boot's auto-configuration simplifies development, but it can also hide internal configurations. When something goes wrong, debugging becomes difficult because developers may not clearly understand what Spring Boot is configuring automatically.

- **High Memory Consumption**
  Spring Boot applications require more memory due to the use of embedded servers, JVM, and multiple dependencies. This can be a challenge for systems with limited hardware resources.

- **Slower Startup Time**
  Compared to lightweight frameworks, Spring Boot applications may take longer to start because of class loading, dependency scanning, and auto-configuration processes.

- **Dependency Management Issues**
  Spring Boot uses starter dependencies, which sometimes include unnecessary libraries. This can increase application size and may cause version conflicts if not managed carefully.

- **Difficult Customization**
  While Spring Boot is easy to use for standard applications, customizing default behavior can be complex. Developers need deep knowledge of Spring internals to override or fine-tune configurations.

- **Not Ideal for Simple Applications**
  For very small or simple applications, Spring Boot may feel heavy and over-engineered compared to scripting languages or lightweight frameworks.

- **Debugging and Error Tracing Complexity**
  Error messages and stack traces in Spring Boot can be long and complex, making debugging difficult, especially for beginners.

- **Version Compatibility Issues**
  Spring Boot versions are tightly coupled with specific Spring Framework and Java versions. Upgrading may require careful compatibility checks.

- **Requires Strong Java Knowledge**
  To use Spring Boot effectively, developers must have a solid understanding of core Java concepts, which can be challenging for beginners.

6.8 Applications of Java Spring Boot

Java Spring Boot is widely used in modern software development because it simplifies Java application creation while providing scalability, security, and reliability. It is especially preferred for building production-ready backend systems and enterprise applications due to its embedded servers, auto-configuration, and strong integration support.

- **Web Applications**
  Spring Boot is extensively used to develop dynamic and scalable web applications. It supports Spring MVC and embedded servers, enabling applications to run independently without external deployment.

- **RESTful API Development**
  Spring Boot is commonly used to build REST APIs that act as backend services for web and mobile applications. It supports JSON-based communication and easy request–response handling.

- **Microservices Architecture**
  Spring Boot is one of the most popular frameworks for micro services development. It allows applications to be divided into small, independent services that can be deployed and scaled separately.

- **Enterprise Applications**
  Many large organizations use Spring Boot for enterprise systems due to its stability, transaction management, security, and seamless integration with databases and messaging systems.

- **Database-Driven Applications**
  Spring Boot supports JPA, Hibernate, and JDBC, making it ideal for applications that require secure and efficient data storage and retrieval.

- **Cloud-Based Applications**
  Spring Boot applications can be easily deployed on cloud platforms and work well with container technologies such as Docker, making them suitable for cloud-native development.

- **E-Commerce Applications**
  Spring Boot is used in e-commerce systems for handling user management, product catalogs, orders, payments, and inventory management.

- **Mobile Application Backends**
  Spring Boot acts as a reliable backend for Android and other mobile applications by providing secure APIs and business logic processing.

- **Banking and Financial Systems**
  Due to strong security, scalability, and transaction management, Spring Boot is widely used in banking and financial applications.

- **Workflow and Automation Systems**
  Spring Boot is well-suited for building workflow-based systems such as service request management, approval systems, and automation platforms.

How to Start Learning Java Spring Boot?

To start learning Java Spring Boot, it is important to follow a structured learning approach because Spring Boot is built on top of core Java and the Spring Framework. A strong foundation in Java is essential before moving on to Spring Boot development.

The first step is to learn Core Java concepts. This includes understanding Java syntax, data types, control statements, object-oriented programming principles, exception handling, collections, and basic multithreading. These concepts form the backbone of Spring Boot applications and help in understanding how the framework works internally.

After mastering core Java, the next step is to learn the basics of the Spring Framework. This involves understanding key concepts such as Dependency Injection (DI), Inversion of Control (IoC), beans, and annotations. Learning how Spring manages components and dependencies makes it easier to understand how Spring Boot simplifies configuration and development.

Once Spring fundamentals are clear, learners can move on to Spring Boot basics. This includes learning about auto-configuration, starter dependencies, embedded servers, and application properties. Creating a simple Spring Boot project using Spring Initializr helps learners understand the project structure and application startup process.

The next stage is to focus on web development using Spring Boot. Learners should practice building web applications and RESTful APIs using Spring MVC. Understanding controllers, request mapping, request–response handling, and JSON data exchange is essential for real-world applications.

After web development, learners should learn database integration with Spring Boot. This involves connecting applications to databases using JPA, Hibernate, or JDBC and performing CRUD operations. Database integration is crucial for building data-driven applications.

Finally, learners should explore advanced Spring Boot concepts such as Spring Security, exception handling, logging, testing, and deployment. Implementing a small project, such as a service request or workflow automation system, helps apply concepts practically and builds confidence in Spring Boot development.

Advantages of Java Spring Boot :

1. **Easy and Simple Development**
   Spring Boot reduces the complexity of Java application development by providing auto-configuration and starter dependencies. This minimizes manual configuration and allows developers to focus on business logic.

2. **Rapid Application Development**
   With embedded servers and predefined configurations, Spring Boot enables faster development, testing, and deployment of applications.

3. **Standalone Applications**
   Spring Boot applications can run independently using embedded servers like Tomcat, eliminating the need for external application servers.

4. **Auto-Configuration Feature**
   Spring Boot automatically configures application components based on the dependencies used, reducing boilerplate code and setup time.

5. **Microservices Friendly**
   Spring Boot is widely used for building micro services due to its lightweight nature, fast development cycle, and easy integration with REST APIs.

6. **Strong Database Integration**
   It provides seamless integration with databases using JPA, Hibernate, and JDBC, making data management efficient and reliable.

7. **Built-in Security Support**
   Spring Boot integrates easily with Spring Security to implement authentication, authorization, and role-based access control.

8. **Scalability and Performance**
   Spring Boot applications are highly scalable and can handle large numbers of users and transactions efficiently.

9. **Easy Maintenance and Monitoring**
   Features like Spring Boot Actuator help in monitoring application health, performance, and metrics in production environments.

10. **Strong Community and Long-Term Support**
    Backed by a large developer community and continuous updates, Spring Boot ensures reliability and long-term stability.

Disadvantages of Java Spring Boot:-

1. **High Memory Consumption**
   Spring Boot applications consume more memory because they run on the Java Virtual Machine (JVM) and include embedded servers along with multiple dependencies.
2. **Slower Startup Time**
   Due to auto-configuration, class path scanning, and dependency loading, Spring Boot applications may take longer to start compared to lightweight frameworks.
3. **Steep Learning Curve**
   Beginners may find Spring Boot difficult to learn as it requires a solid understanding of core Java and Spring concepts such as Dependency Injection and annotations.
4. **Hidden Auto-Configuration Complexity**
   While auto-configuration reduces setup effort, it can make debugging difficult because developers may not clearly understand which configurations are applied internally.
5. **Dependency Management Issues**
   Starter dependencies may include unnecessary libraries, increasing application size and sometimes causing version conflicts if not managed properly.
6. **Not Suitable for Small Applications**
   For very simple or small-scale applications, Spring Boot can feel heavy and over-engineered.
7. **Complex Debugging**
   Error logs and stack traces in Spring Boot can be long and complex, making it harder to identify the root cause of issues.
8. **Performance Overhead**
   Spring Boot may not be ideal for low-level or real-time applications due to framework overhead and garbage collection pauses.
9. **Version Compatibility Constraints**
   Spring Boot versions depend on specific Java and Spring versions, which can create challenges during upgrades.
10. **Requires Strong Java Knowledge**
    Effective use of Spring Boot requires good understanding of Java and Spring internals, which can be challenging for beginners.

6.9 Java Development Steps

Java development follows a systematic process to design, build, test, and deploy applications efficiently. The following steps explain the Java development process in a clear and structured manner, suitable for project documentation and exams.

1. **Requirement Analysis**
   The first step in Java development is understanding the application requirements. This includes identifying the problem to be solved, defining system functionalities, and gathering user needs. Proper requirement analysis helps in planning the application structure.
2. **Design Phase**
   In this step, the overall design of the application is prepared. This includes designing classes, objects, data flow, and system architecture. UML diagrams such as class diagrams and sequence diagrams are often used during this phase.
3. **Environment Setup**
   The Java Development Kit (JDK) is installed, and environment variables are configured. An Integrated Development Environment (IDE) such as Eclipse or IntelliJ IDEA is set up to write and manage Java code efficiently.
4. **Coding / Implementation**
   During this phase, the actual Java code is written based on the design. Developers create classes, methods, and packages using object-oriented programming concepts. Proper coding standards are followed to ensure readability and maintainability.
5. **Compilation**
   The written Java source code is compiled using the Java compiler. The compiler converts the source code into bytecode, which can be executed by the Java Virtual Machine (JVM).
6. **Execution**
   The compiled bytecode is executed on the JVM. The JVM ensures platform independence by running the same bytecode on different operating systems.
7. **Testing and Debugging**
   The application is tested to identify and fix errors. Unit testing, integration testing, and debugging are performed to ensure that the application functions correctly and meets requirements.
8. **Deployment**
   After successful testing, the Java application is deployed on the target environment such as a server or cloud platform. Deployment makes the application available for users.
9. **Maintenance and Enhancement**
   In the final step, the application is maintained by fixing bugs, improving performance, and adding new features based on user feedback.

Java

Java is a high-level, object-oriented, and platform-independent programming language widely used for developing reliable, secure, and scalable software applications. It was developed by James Gosling at Sun Microsystems and officially released in 1995. Java was designed with the primary goal of allowing developers to write programs that can run on any computing platform without modification, which led to its well-known principle, "Write Once, Run Anywhere."

Java programs are written in a human-readable source code format and compiled into an intermediate form known as bytecode. This bytecode is executed by the Java Virtual Machine (JVM), which acts as an abstraction layer between the program and the operating system. Because of the JVM, Java applications can run on different operating systems such as Windows, Linux, and macOS, as long as a compatible JVM is available. This feature makes Java highly portable compared to many other programming languages.

Java follows the object-oriented programming (OOP) paradigm, which helps in building modular, reusable, and maintainable applications. It supports key OOP concepts such as encapsulation, inheritance, abstraction, and polymorphism. These principles allow developers to organize code efficiently, reduce complexity, and manage large software systems effectively. Java also enforces strong typing and compile-time checking, which helps detect errors early in the development process.

One of the major strengths of Java is its automatic memory management. Java uses a garbage collection mechanism that automatically allocates and deallocates memory, reducing the chances of memory leaks and system crashes. This feature relieves developers from manual memory management and improves application stability. In addition, Java provides a robust exception handling mechanism that helps manage runtime errors gracefully.

CHAPTER – 7

MODULES USED IN PROJECT

1. Admin Module

The Admin Module is the central control unit of the Service Request & Workflow Automation System. This module allows administrators to manage users, define service request categories, set request priorities, and configure workflow rules. Administrators can view all service requests raised by users and monitor their progress at different stages. This module ensures proper control, accountability, and smooth functioning of the system by providing centralized administration and decision-making capabilities.

2. User Registration & Login Module

The User Registration & Login Module provides secure access to the system for authorized users. New users can register by submitting valid details, while existing users can log in using their credentials. This module ensures authentication and role-based access control, allowing users to access only permitted features. It enhances system security, protects user data, and prevents unauthorized access.

3. Service Request Module

The Service Request Module enables users to create and submit service requests by providing necessary details such as request type, description, and priority. Each request is assigned a unique request ID and stored in the database. Users can view their submitted requests and track their current status. This module forms the core functionality of the system by enabling structured and efficient service request submission.

4. Workflow Automation Module

The Workflow Automation Module manages the flow of service requests through predefined approval and processing stages. Based on configured rules, service requests are automatically routed to the appropriate authorities or departments. This module eliminates manual forwarding, reduces processing delays, and ensures consistency in request handling. Workflow automation improves efficiency, transparency, and reliability within the system.

5. Task Assignment & Tracking Module

The Task Assignment & Tracking Module allows administrators to assign approved service requests to appropriate teams or service personnel. This module tracks the progress of assigned tasks in real time and ensures timely completion of requests. It improves workload distribution, accountability, and monitoring of service activities across the organization.

6. Notification Module

The Notification Module is responsible for sending alerts and updates to users and administrators at various stages of the service request lifecycle. Notifications are triggered during request creation, approval, status updates, and completion. This module improves

communication, reduces manual follow-ups, and keeps all stakeholders informed about request progress.

7. Report & Analytics Module

The Report & Analytics Module generates detailed reports related to service request status, resolution time, and team performance. These reports help administrators analyze system efficiency and identify areas for improvement. This module supports data-driven decision-making and enhances overall service quality within the organization.

7.1 Install Java Step-by-Step in Windows or MAC

Java is a versatile programming language widely used for building applications. To start coding in Java, you first need to install the Java Development Kit (JDK) on your system. This article provides detailed steps for installing Java on Windows 7, 8, 10, 11, Linux Ubuntu, and macOS.

7.2 How to Install Java on Windows and Mac

Here, we will discuss how to download and install Java on a Windows 64-bit Machine and set up the environment to run the first Java program on the command prompt. The process is the same for all Windows i.e. 7, 8, 10, and 11.

Step-1: Visit the official Oracle website https://www.oracle.com/in/java/. Click on the "Download Java" icon as shown in the below image.

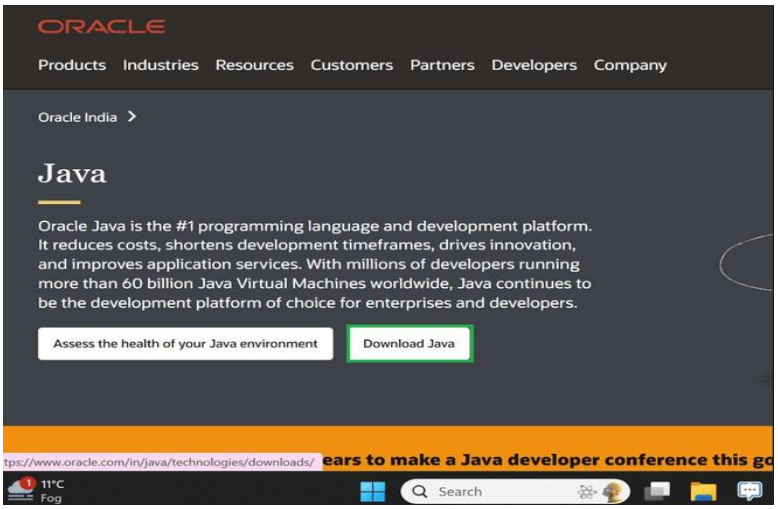


Fig 7.1 Open Java website to download the Java latest version

Step-2: Now you can see the latest version is JDK 23 and there are options for Linux, macOS, and Windows. Click on the Windows option and then click the x64 Installer option to download the .exe file for 64-bit Windows OS

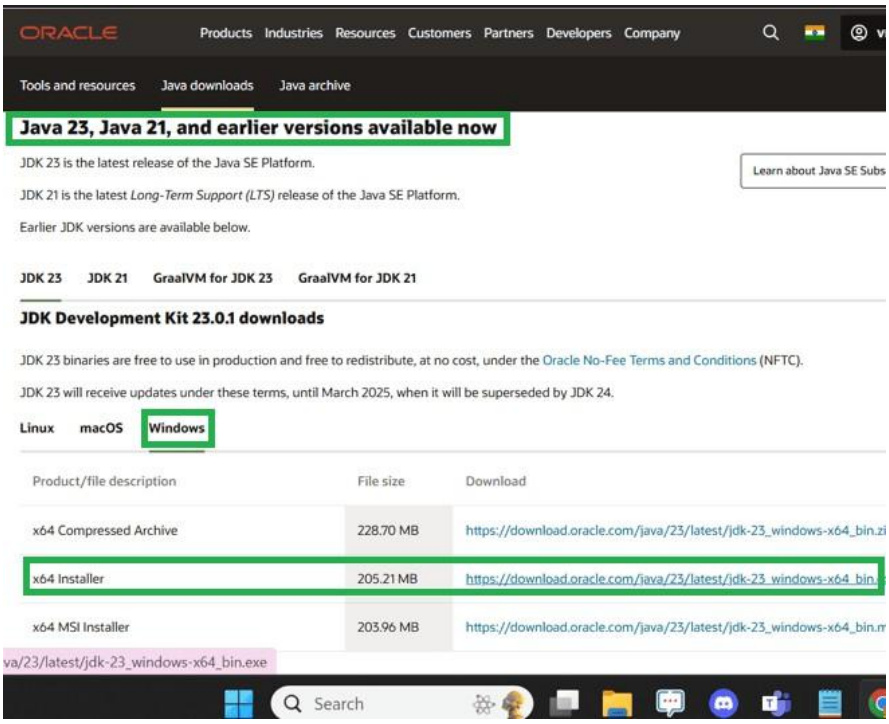


Fig 7.2 Click on Download

You can simply use this link to download Java for Windows 64 Bit. After successfully downloading JDK 23, now we need to install the Java on our Windows System.

Step-3: Go to Downloads and double-click on that downloaded jdk-23_windows-x64_bin.exe file. So now Java Installation Wizard get open and then click Next button

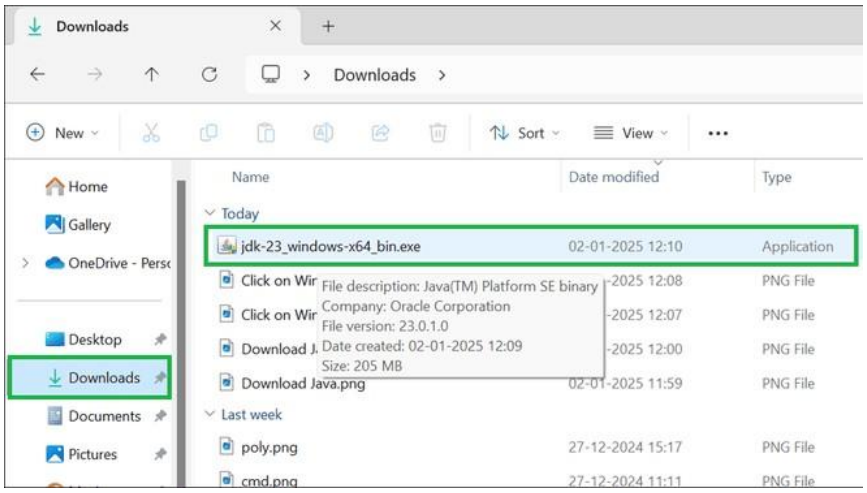


Fig 7.3 Java Development Kit (JDK) Setup File

Again click on the Next button to install JDK.

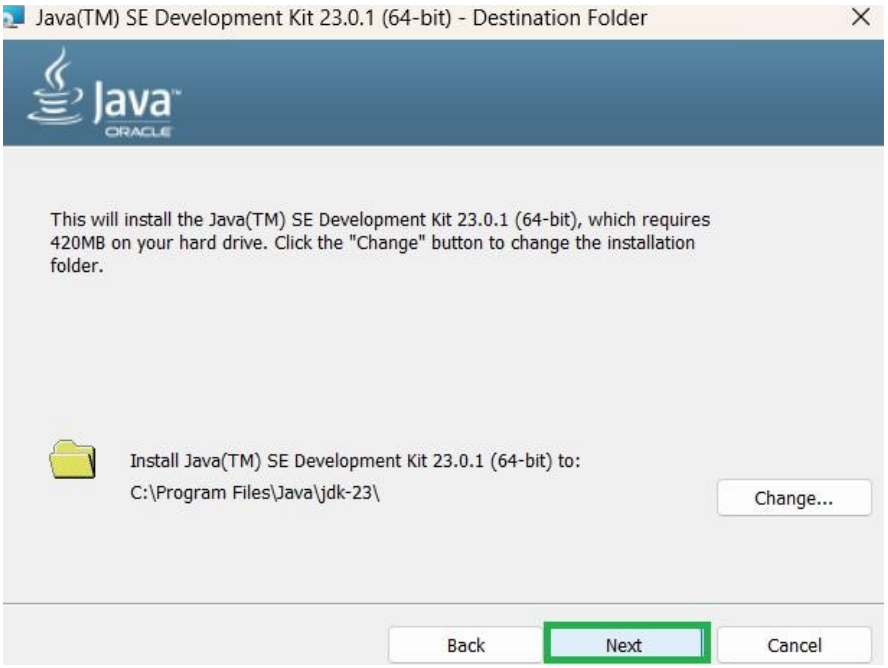


Fig 7.4 JDK Installation Path Configuration

The installation will begin as shown below:

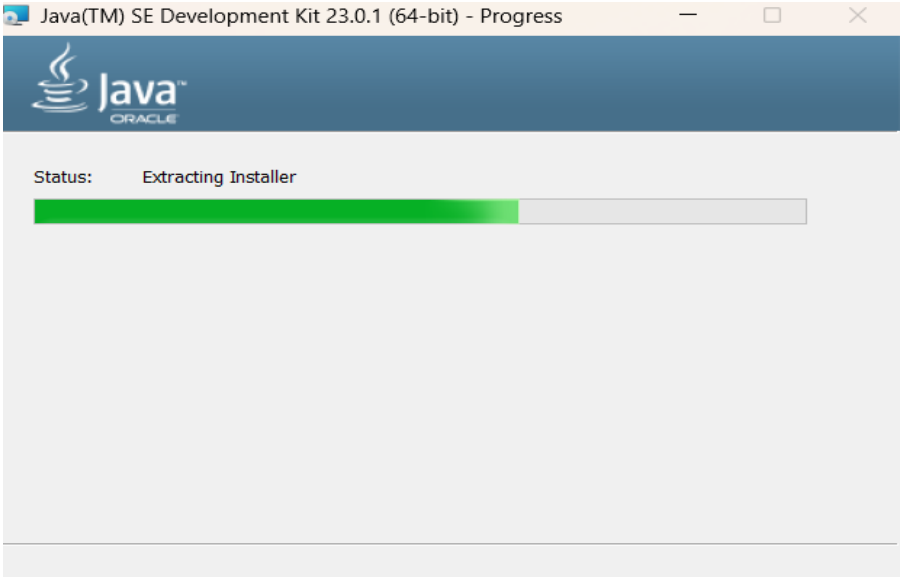


Fig 7.5 JDK Setup Process in Progress

The Java installation is successfully completed as shown below. Now click on "close" to finish.

Fig 7.6 Successful Installation of Java Development Kit (JDK)

Step-4: Now we will set the environment variables for Windows OS. Open the C drive, go to Program Files > Java > jdk-23 (or your installed version) > bin folder. Now, copy the path and we will use this path when configuring the environment variables.

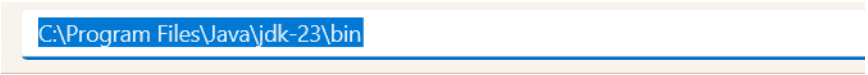


Fig 7.7 Java Environment Path Configuration

Search for environment variable on your system, then click on the Environment Variables button.

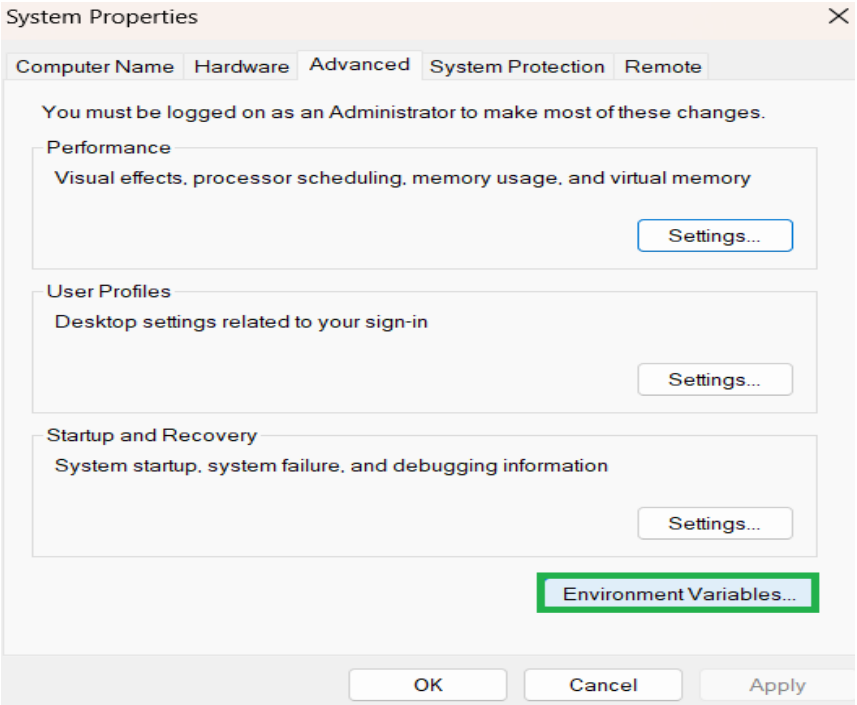


Fig 7.8 Accessing Environment Variables in System Properties

In the system variable section, select the path variable and then click the option Edit.

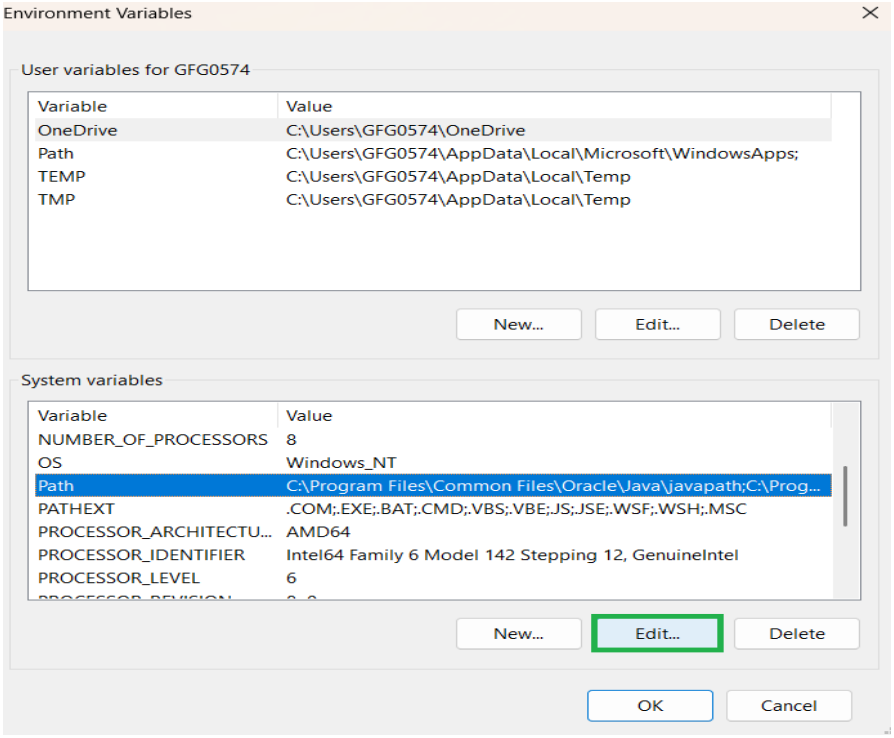


Fig 7.9 Environment Variables Configuration for Java

Paste the path that copied earlier, and click OK to save the changes.

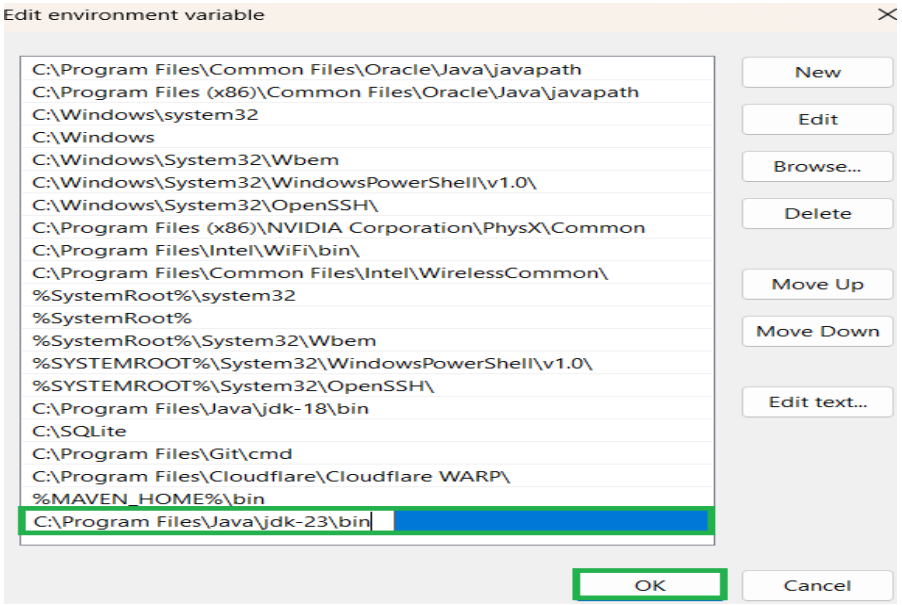


Fig 7.10 Configuration of System PATH Environment Variable

Step-5: Verify the Installation. Open command prompt and type the below command to verify the Java version that is installed in the system.

Java –version

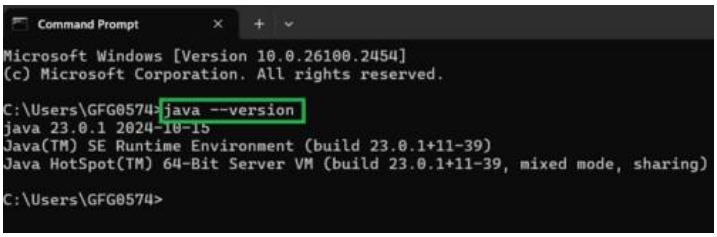


Fig 7.11 Verifying Installed Java Version via Command Line

Now let us run our first Java program, to do this, follow the below steps:

Step 1: Open any Drive, and in the drive, create a text file and then change the file name with .java extension as shown below:

File.java

Step 2: Now, write a simple Java code in notepad.

- Right-click on the file.java and then select open with > Notepad.
- Inside the file, write the Java code.

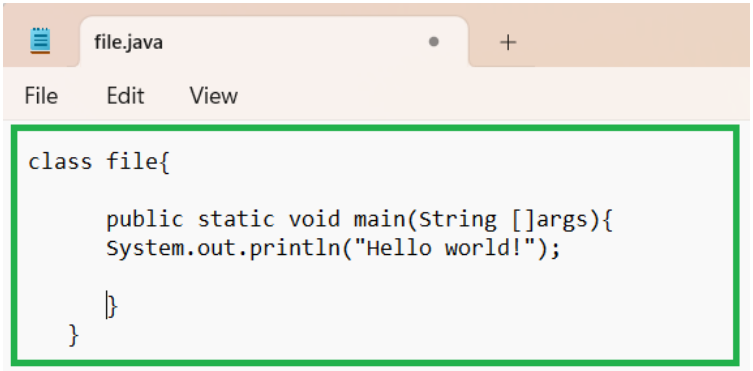


Fig 7.12 Basic Java Program to Print "Hello World"

Important Point: In Java, the class name should be equivalent to the file name. In this case, the file name is file.java, so the class name inside the file should be file.

Step 3: Now, go to the directory where your Java file is saved.

Open command prompt and navigate to the D drive, where the file.java is located. To do this, type "dir" as shown below:

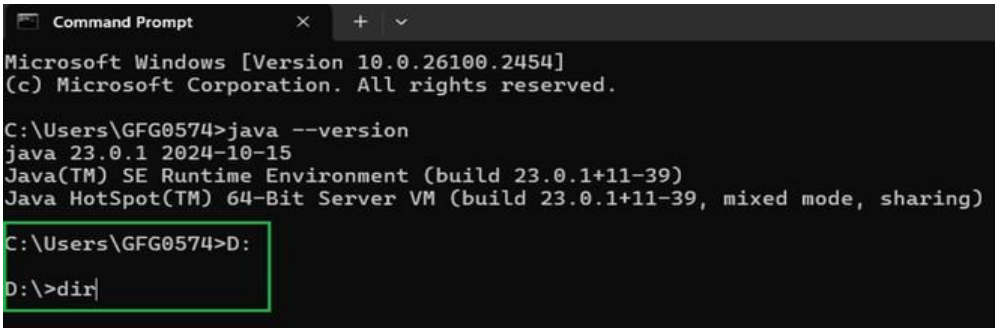


Fig 7.13 Command-Line Verification of Java Installation and Drive Navigation

Step 4: The dir command display all the contents of the directory.

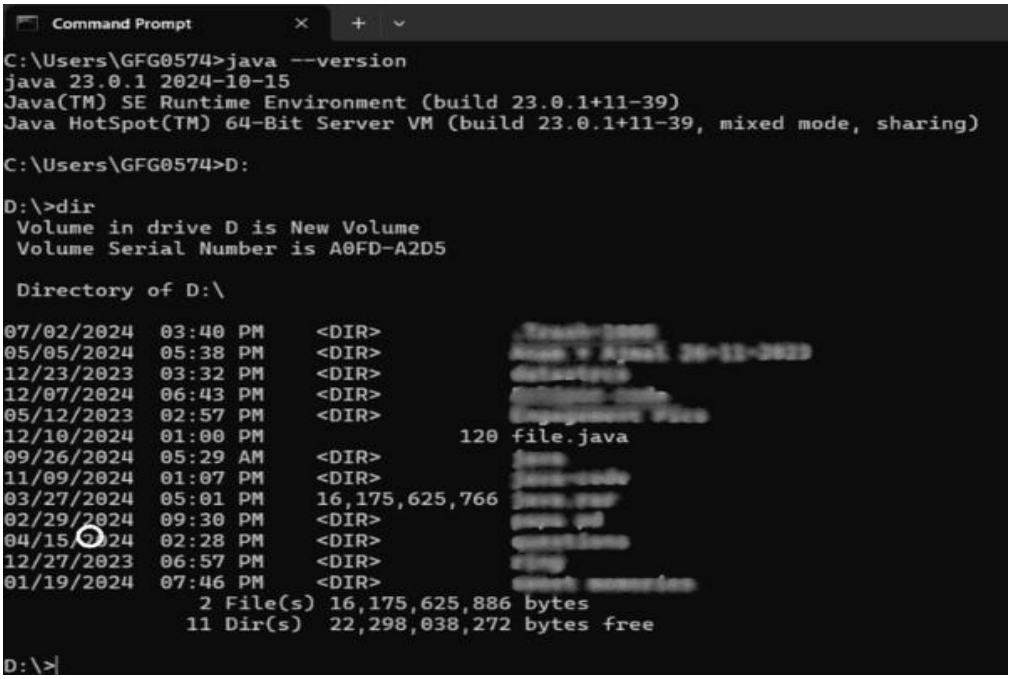


Fig 7.14 Directory Listing in Windows Command Prompt

Step 5: Compile the Java program. The javac command is used to compile the Java program.

Syntax:

javac filename.java

To run the program use the below command:

java filename

Here, in this case the filename is file.

```
C:\Users\GFG0574>D:

D:\>dir
 Volume in drive D is New Volume
 Volume Serial Number is A0FD-A2D5

 Directory of D:\

07/02/2024  03:40 PM    <DIR>          .Trash-1000
05/05/2024  05:38 PM    <DIR>          [illegible]
12/23/2023  03:32 PM    <DIR>          [illegible]
12/07/2024  06:43 PM    <DIR>          [illegible]
05/12/2023  02:57 PM    <DIR>          [illegible]
12/10/2024  01:00 PM               120 file.java
09/26/2024  05:29 AM    <DIR>          [illegible]
11/09/2024  01:07 PM    <DIR>          [illegible]
03/27/2024  05:01 PM    16,175,625,766 [illegible]
02/29/2024  09:30 PM    <DIR>          [illegible]
04/15/2024  02:28 PM    <DIR>          [illegible]
12/27/2023  06:57 PM    <DIR>          [illegible]
01/19/2024  07:46 PM    <DIR>          [illegible]
               2 File(s) 16,175,625,886 bytes
              11 Dir(s)  22,298,038,272 bytes free

D:\>javac file.java

D:\>java file
Hello world!

D:\>
```

Fig 7.15 Compilation and Execution of a Java Program via Command Line

We have successfully run our first Java program on Windows OS.

Download and Install Java - Linux

Step 1: Visit the official oracle website https://www.oracle.com/in/java/.

- You will need to accept the terms of Oracle's license agreement before downloading.
- On the download page, accept the Oracle Binary Code License Agreement for Java SE.
- Select the Linux version.
- If you are using a standard 64-bit system, select the file for x64 architecture. If you are using an ARM-based system, select the file for arm64.
- Download the. tar.gz file for Linux.

  This is the format used for installing Java on most Linux systems.

Step 2: Now, open the terminal on your Linux system and navigate to the folder where you downloaded the file.

cd ~/Downloads

Use the below command to extract the contents of the .tar.gz file.

tar -xvzf jdk-23_linux-x64_bin.tar.gz

Move the extracted folder to the /opt directory:

sudo mv jdk-23 /opt/

Step 3: After Installation, you need to set up the environment variable for Linux.

Open the terminal and type the below command to open the shell configuration file.

nano ~/.bashrc

Add the following lines at the end of the file.

export JAVA_HOME=/opt/jdk-23

export PATH=$JAVA_HOME/bin:$PATH

Apply the changes by running the below command:

source ~/.bashrc

Step 4: Now, verify the installation on Linux by running the below command:

java --version

The installed Java version on Linux will be displayed.

Download and Install Java - macOS

Step 1: Visit the official oracle website https://www.oracle.com/in/java/. Select the macOS version and download the .dmg file.

Step 2: Now open the .dmg file and double-click the installer package (.pkg) to begin the installation process. Then follow the instructions to complete the installation.

Step 3: Setting the environment variable on macOS is optional, but it is recommended, if you want to use Java for development purposes (e.g., running Java- based applications, using tools like Maven or Gradle).

To set the environment variable foe macOS, open the Terminal and edit the shell configuration file (e.g., .zshrc):

nano ~/.zshrc

Add the following lines at the end of the file to set the Java_Home.

export JAVA_HOME=$(/usr/libexec/java_home)

export PATH=$JAVA_HOME/bin:$PATH

Save and close. To apply the changes, run the following command:

source ~/.zshrc

After these steps, Java should be installed and ready to use on your macOS system.

CHAPTER – 8

TESTING

8.1 System Test

The purpose of system testing is to identify errors and ensure that the developed Service Request & Workflow Automation System functions according to the specified requirements. Testing is the process of executing a software system to detect defects, verify correctness, and validate that the system meets user expectations.

System testing evaluates the complete and integrated application to confirm that all modules such as user registration, login, service request creation, workflow processing, task assignment, and status tracking work properly. The goal is to ensure that the system operates reliably, efficiently, and without failure under normal and abnormal conditions. Various testing techniques are used to verify the performance, security, and usability of the system.

8.2 Types Of Tests

8.2.1 Unit testing

Unit testing involves testing individual modules of the Service Request & Workflow Automation System to verify that each component functions correctly in isolation. Each unit, such as login validation, user registration, service request submission, and password change, is tested independently.

This testing ensures that internal logic, decision branches, and data validations produce correct outputs for given inputs. Unit testing is performed after the completion of each module and before integration. It helps in detecting errors early and improves code quality.

8.2.2 Integrated Testing

Integration testing is performed after unit testing to verify that multiple modules of the system work together correctly. In this project, integration testing ensures smooth interaction between modules such as user management, service request handling, workflow processing, and database connectivity.

This testing identifies issues related to data flow, interface mismatches, and communication between integrated components. It confirms that individually tested modules function correctly when combined as a complete system.

8.2.3 Functional test

Functional testing ensures that all features of the Service Request & Workflow Automation System work according to the functional requirements and system specifications.

Functional testing is centered on the following items:

1. **Valid Inputs:** Accepting valid user data such as login credentials and service request details.
2. **Invalid Inputs:** Rejecting invalid or incomplete inputs.
3. **Functions:** Verifying core functions like request creation, assignment, approval, and status updates.
4. **Outputs:** Displaying correct messages, notifications, and request status.
5. **System Procedures:** Ensuring proper workflow execution and system responses.

This testing validates the complete business flow of service request processing.

8.2.4 White Box Testing

White Box Testing is conducted with full knowledge of the system's internal structure and source code. It verifies logical paths, loops, conditions, and internal operations of the application.

In this project, white box testing is used to test backend services, database interactions, and workflow logic that are not visible through the user interface.

8.2.5 Black Box Testing

Black Box Testing evaluates the system without any knowledge of internal code structure. The tester provides inputs and checks outputs based solely on functional requirements.

For the Service Request & Workflow Automation System, black box testing is used to test user login, registration, service request submission, and status tracking without considering how the system processes data internally.

8.2.6 Unit Testing

- Unit testing is conducted during the development phase.
- Functional tests are executed manually.
- Integration testing ensures seamless interaction between system modules.

Test Objectives:

- All form fields must validate properly.
- Pages must load correctly from defined links.
- Responses and messages should be displayed without delay.

Features Tested:

- Proper data format validation.
- Prevention of duplicate entries.
- Correct navigation through all links.

Acceptance Testing

User Acceptance Testing (UAT) is performed to ensure that the system meets user expectations and functional requirements. End users test the system in a real-time environment to verify usability, accuracy, and workflow performance.

Test Results: All acceptance test cases were executed successfully with no defects detected.

Test Cases

Test Case – 1: Login Module

Function: Login
Expected Result:
The system should validate user credentials and verify user existence in the database.
Actual Result:
User credentials are successfully validated against the database.
Priority: High

Test Case – 2: User Registration Module

Function: User Registration
Expected Result:
The system should validate all input fields and store user details in the database.
Actual Result:
All input fields are validated and user data is successfully saved.
Priority: High

Test Case – 3: Change Password Module

Function: Change Password
Expected Result:
If the old password does not match, an error message should be displayed.
Actual Result:
The system correctly validates old and new passwords and displays appropriate messages.
Priority: High

CHAPTER - 9

IMPLEMENTATION

9.1 Methodology

The Service Request & Workflow Automation System follows a web-based and workflow-driven methodology to automate the process of creating, managing, and resolving service requests within an organization. The methodology is designed to reduce manual effort, improve transparency, and ensure timely processing of service requests. The overall implementation is divided into multiple phases, including requirement analysis, system design, backend development, database integration, workflow automation, and result monitoring.

1. Requirement Analysis and System Planning

In the initial phase, system requirements are gathered by analyzing existing manual service request processes. Common issues such as delayed approvals, lack of tracking, and poor communication are identified. Based on these requirements, functional modules such as user management, service request handling, workflow automation, task assignment, and reporting are defined.

2. System Design and Architecture

The system is designed using a layered architecture, which includes the presentation layer, business logic layer, and data access layer. UML diagrams such as use case diagrams, class diagrams, and sequence diagrams are prepared to represent system flow and interactions. This design approach ensures modularity, scalability, and ease of maintenance.

3. Backend Development Using Spring Boot

The backend of the application is developed using Java Spring Boot. RESTful APIs are created to handle user authentication, service request creation, workflow transitions, and task assignments. The business logic layer processes workflow rules and request status updates, ensuring smooth automation of service request handling.

4. Database Integration

The system uses MySQL as the backend database to store user information, service request details, workflow stages, task assignments, and status updates. JPA/Hibernate is used for object-relational mapping, allowing efficient interaction between Java objects and database tables while maintaining data consistency.

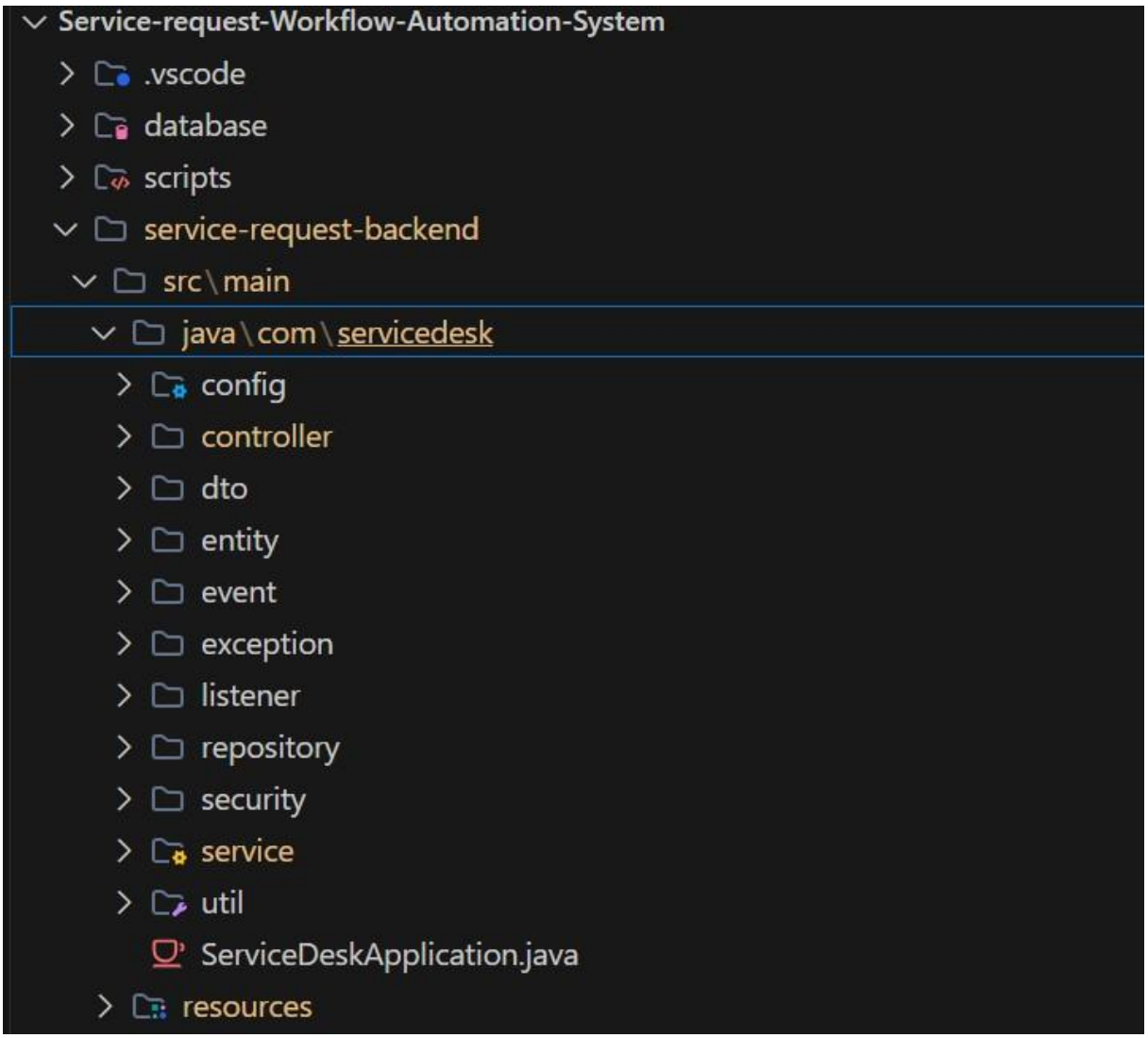
5. Workflow Automation and Task Assignment

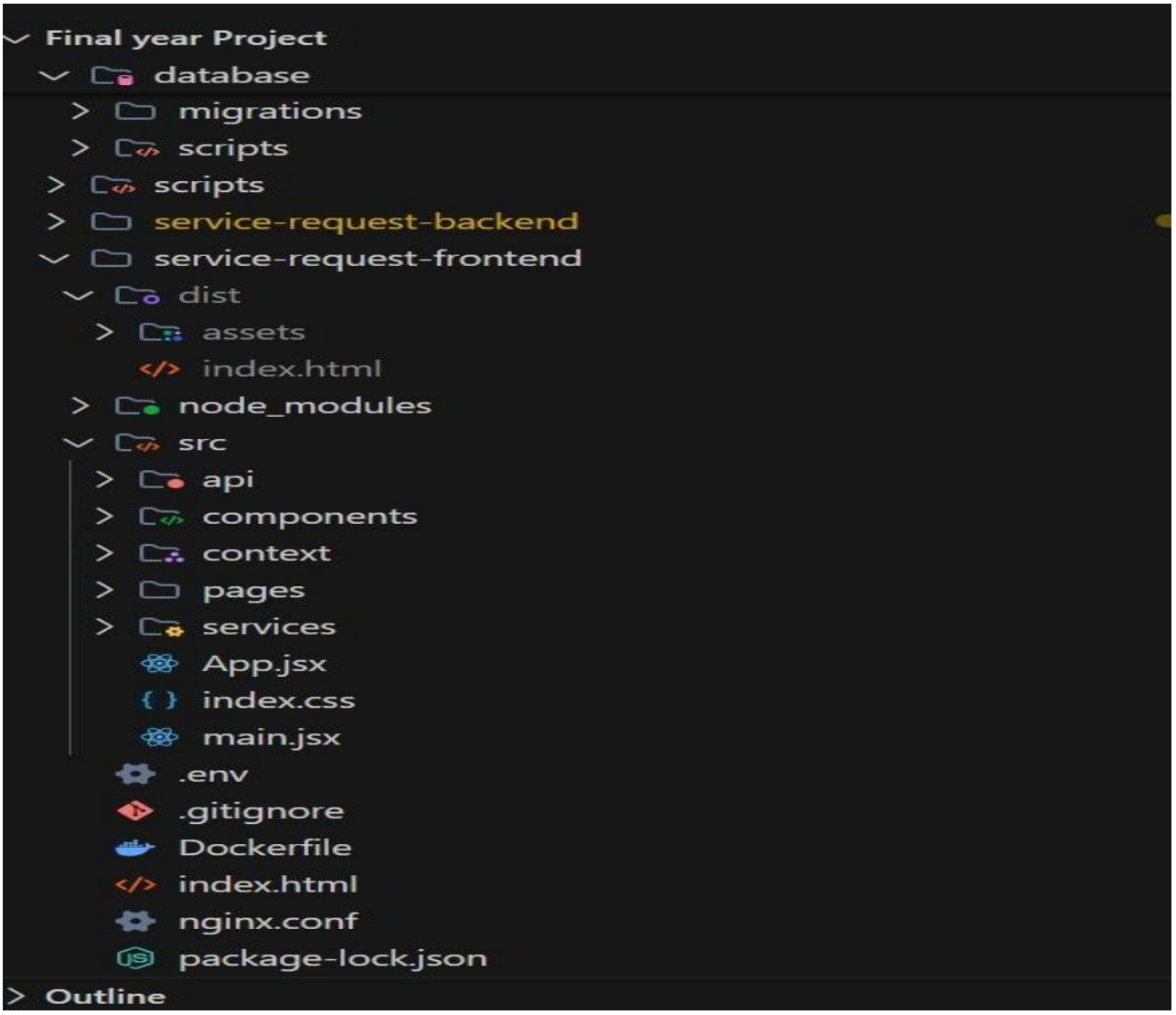
Once a service request is submitted, it is automatically routed through predefined workflow stages such as request submission, approval, task assignment, processing, and completion. Workflow rules determine request priority and responsible personnel. This automation minimizes manual intervention and ensures timely resolution of requests.

6. Notification and Status Tracking

The system includes a notification mechanism to inform users and administrators about important events such as request submission, approval, assignment, and completion. Users can track the real-time status of their service requests through the dashboard, improving transparency and communication.

PROJECT STRUCTURE





9.2 Source Code

```
package com.servicedesk.controller;

import com.servicedesk.dto.ApiResponse;
import  com.servicedesk.dto.ServiceCategoryDTO;
import com.servicedesk.service.ServiceCatalogService;
import jakarta.validation.Valid;
import  org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.web.bind.annotation.*;
```

```
import java.util.List;

/**
 * Admin Category Controller
 * Provides full CRUD operations for service categories (admin only)
 */
@RestController
@RequestMapping("/admin/categories")
@CrossOrigin(origins = "*", maxAge = 3600)
@PreAuthorize("hasRole('ROLE_ADMIN')")
public class AdminCategoryController {

    private static final Logger log =
LoggerFactory.getLogger(AdminCategoryController.class);

    @Autowired
    private ServiceCatalogService serviceCatalogService;

    /**
     * Get all categories (including inactive)
     * GET /api/admin/categories
     */
    @GetMapping
    public ResponseEntity<List<ServiceCategoryDTO>> getAllCategories() {
        log.info("Admin fetching all categories");
        List<ServiceCategoryDTO> categories =
serviceCatalogService.getAllCategories();
        log.info("Found {} categories", categories.size());
        return ResponseEntity.ok(categories);
    }

    /**
     * Get category by ID
     * GET /api/admin/categories/{id}
     */
    @GetMapping("/{id}")
    public ResponseEntity<?> getCategoryById(@PathVariable Long id) {
        try {
            log.info("Admin fetching category with id: {}", id);
            ServiceCategoryDTO category =
serviceCatalogService.getCategoryById(id);
            return ResponseEntity.ok(category);
        } catch (Exception e) {
            log.error("Error fetching category {}: {}", id, e.getMessage());
```

```
            return ResponseEntity.badRequest()
                    .body(new ApiResponse(false, e.getMessage()));
        }
    }

    /**
     * Create new category
     * POST /api/admin/categories
     */
    @PostMapping
    public ResponseEntity<?> createCategory(@Valid @RequestBody ServiceCategoryDTO
dto) {
        try {
            log.info("Admin creating new category: {}", dto.getName());

            // Validate unique name
            if (serviceCatalogService.categoryNameExists(dto.getName())) {
                log.warn("Category name already exists: {}", dto.getName());
                return ResponseEntity.badRequest()
                        .body(new ApiResponse(false, "Category name already
exists"));

            }

            ServiceCategoryDTO created =
serviceCatalogService.createCategory(dto);
            log.info("Category created successfully with id: {}",
created.getId());
            return ResponseEntity.status(HttpStatus.CREATED).body(created);
        } catch (Exception e) {
            log.error("Error creating category: {}", e.getMessage(), e);
            return ResponseEntity.badRequest()
                    .body(new ApiResponse(false, "Failed to create category: " +
e.getMessage()));
        }
    }

    /**
     * Update category
     * PUT /api/admin/categories/{id}
     */
    @PutMapping("/{id}")
    public ResponseEntity<?> updateCategory(
            @PathVariable Long id,
            @Valid @RequestBody ServiceCategoryDTO dto) {
        try {
```

```
            log.info("Admin updating category {}: {}", id, dto.getName());

            // Validate unique name (excluding current category)
            if (serviceCatalogService.categoryNameExistsExcludingId(dto.getName(),
id)) {
                log.warn("Category name already exists: {}", dto.getName());
                return ResponseEntity.badRequest()
                        .body(new ApiResponse(false, "Category name already
exists"));

            }

            ServiceCategoryDTO updated = serviceCatalogService.updateCategory(id,
dto);
            log.info("Category {} updated successfully", id);
            return ResponseEntity.ok(updated);
        } catch (Exception e) {
            log.error("Error updating category {}: {}", id, e.getMessage(), e);
            return ResponseEntity.badRequest()
                    .body(new ApiResponse(false, "Failed to update category: " +
e.getMessage()));
        }
    }

    /**
     * Toggle category active status
     * PATCH /api/admin/categories/{id}/status
     */
    @PatchMapping("/{id}/status")
    public ResponseEntity<?> toggleCategoryStatus(@PathVariable Long id) {
        try {
            log.info("Admin toggling status for category {}", id);
            ServiceCategoryDTO category =
serviceCatalogService.toggleCategoryStatus(id);
            log.info("Category {} status toggled to: {}", id,
category.getIsActive());
            return ResponseEntity.ok(category);
        } catch (Exception e) {
            log.error("Error toggling category {} status: {}", id, e.getMessage(),
e);
            return ResponseEntity.badRequest()
                    .body(new ApiResponse(false, "Failed to toggle category
status: " + e.getMessage()));
        }
    }
```

```
    /**
     * Delete (soft delete) category
     * DELETE /api/admin/categories/{id}
     */
    @DeleteMapping("/{id}")
    public ResponseEntity<?> deleteCategory(@PathVariable Long id) {
        try {
            log.info("Admin deleting category {}", id);
            serviceCatalogService.deleteCategory(id);
            log.info("Category {} deleted successfully", id);
            return ResponseEntity.ok(new ApiResponse(true, "Category disabled
successfully"));
        } catch (Exception e) {
            log.error("Error deleting category {}: {}", id, e.getMessage(), e);
            return ResponseEntity.badRequest()
                    .body(new ApiResponse(false, "Failed to delete category: " +
e.getMessage()));
        }
    }
}
```

CHAPTER – 10

RESULTS

Screenshots:

Service Request & Workflow Automation System

In the proposed project, **Service Request & Workflow Automation System**, the application successfully automates the process of raising, approving, assigning, and tracking service requests within an organization. The system provides a centralized platform where users can submit service requests and administrators can manage workflows efficiently.

The system is implemented using Java Spring Boot, REST APIs, and MySQL, ensuring secure, scalable, and reliable request handling. The application supports role-based access for users and administrators and provides real-time status tracking and notifications.

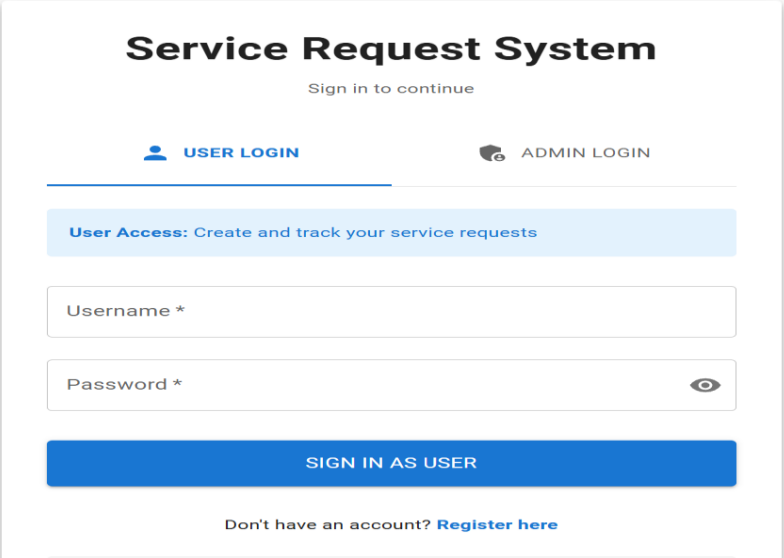


Fig 10.1 User Login Screen

This screen allows registered users to log in to the Service Request System using valid credentials. It provides separate login options for users and administrators. After successful login, users can access their dashboard and service request features.

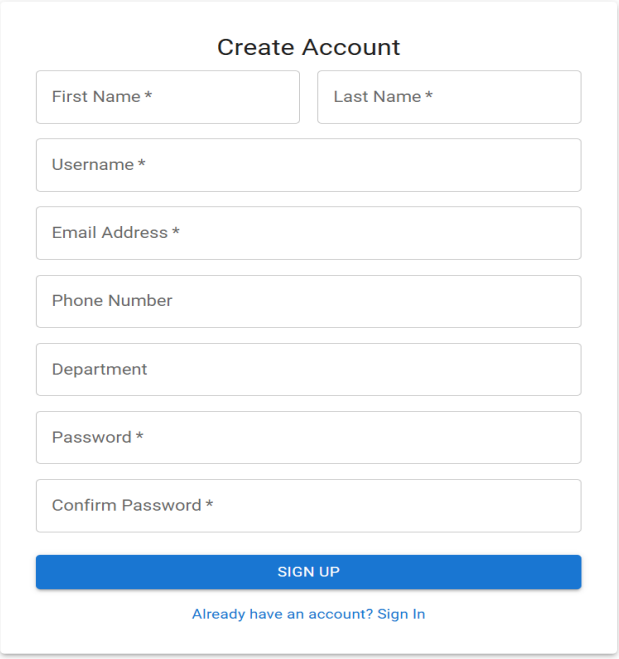


Fig 10.2 User Registration Screen

This screen enables new users to create an account by entering basic personal and login details. Mandatory fields ensure correct data entry and secure registration. Once registered, users can log in and start raising service requests. This module helps manage authorized system users.

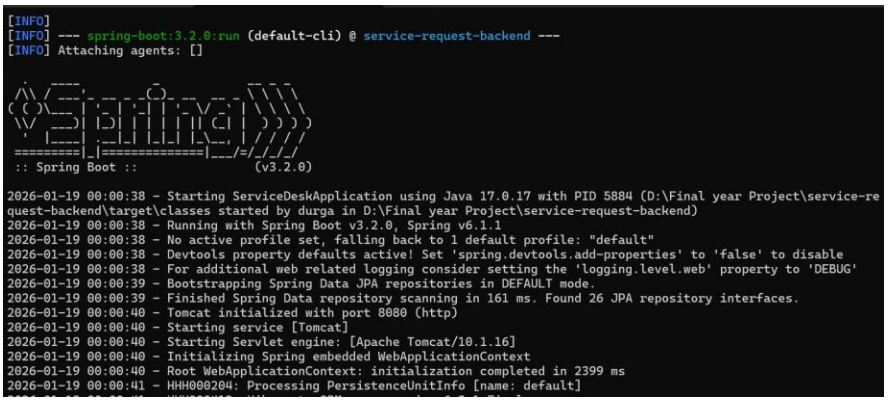


Fig 10.3 Spring Boot Server Startup

Spring Boot backend server for the Service Request & Workflow Automation System. The application runs on Java 17 using an embedded Apache Tomcat server on port 8080. All required components such as JPA repositories and web context are initialized successfully. This confirms that the backend service is ready to handle client requests.

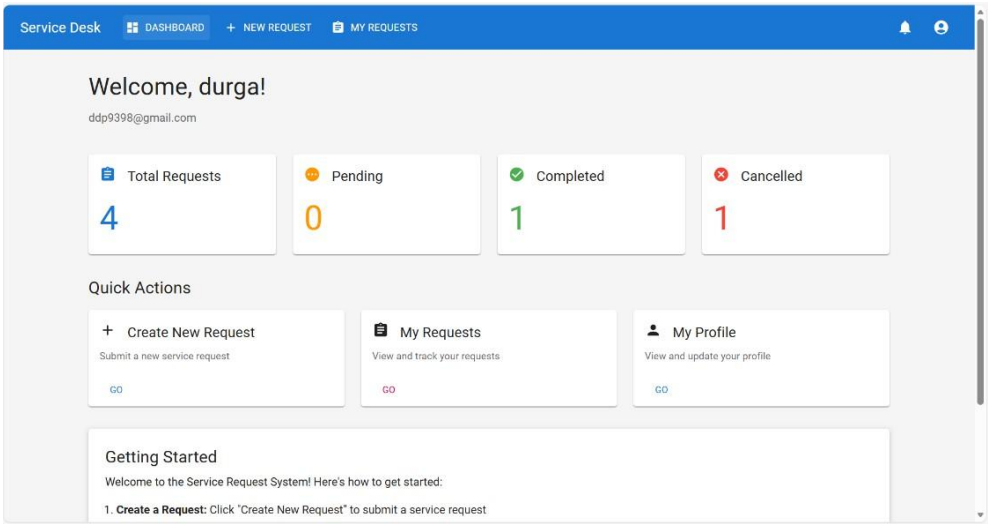


Fig 10.4 User Dashboard

The user dashboard displays an overview of all service requests raised by the user. It shows request statistics such as total, pending, completed, and cancelled requests. Quick action buttons allow users to create new requests and view existing ones easily. This improves user convenience and tracking.

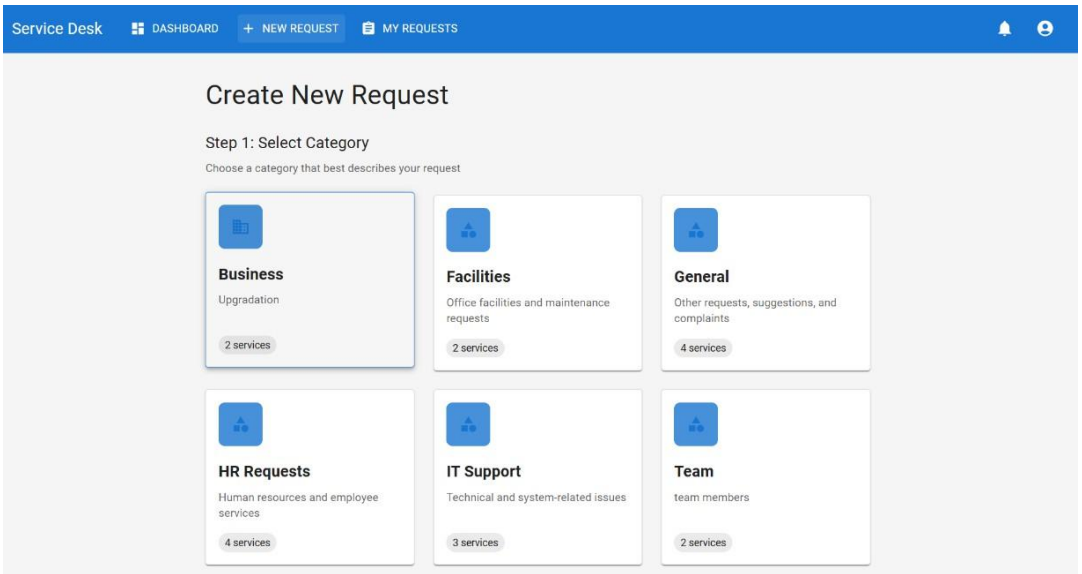


Fig 10.5 Create New Request – Category Selection

Categories help classify requests accurately for faster processing. Once a category is selected, the request is routed to the relevant department. This improves workflow automation and request handling.

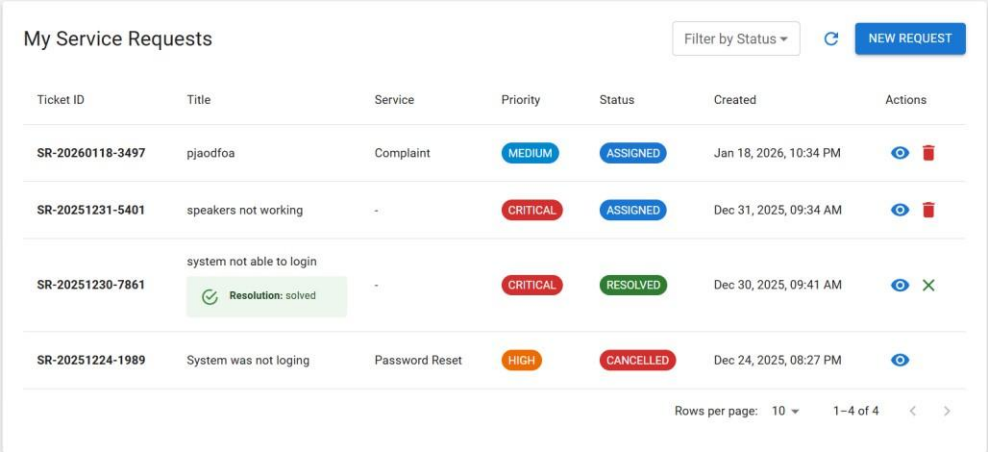


Fig 10.6 My Service Requests

Displays all service requests submitted by the user in a tabular format. It shows request details such as ticket ID, priority, status, and created date. Users can track request progress and view request details. This ensures transparency and easy monitoring.

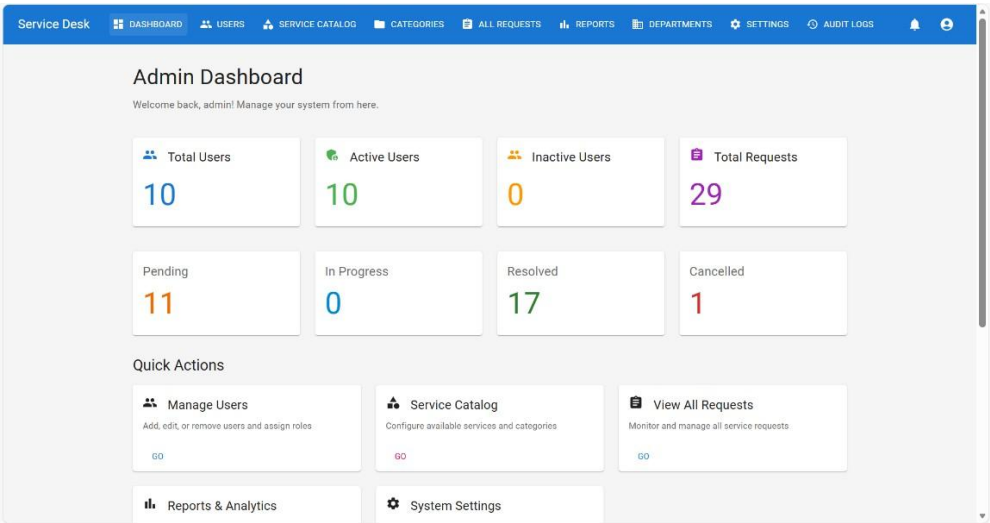
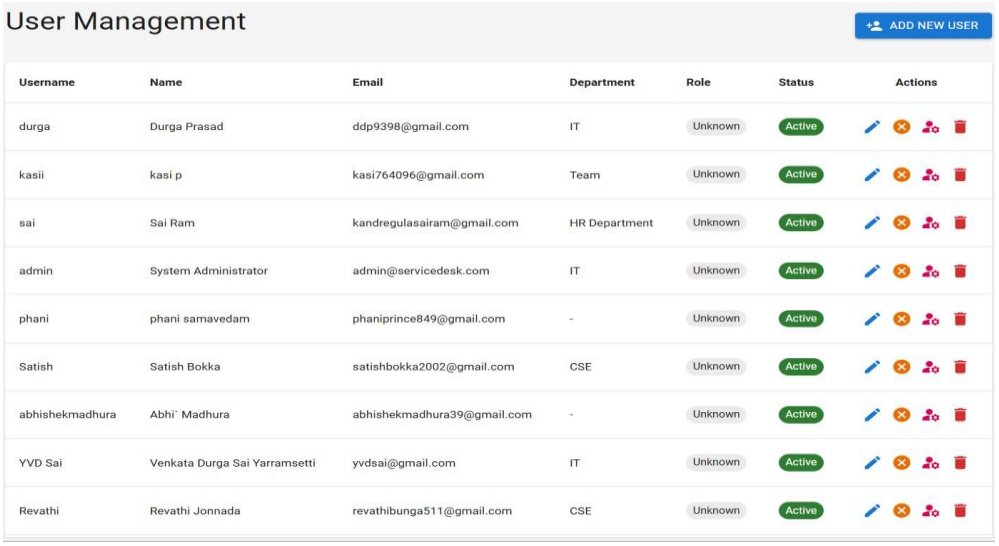


Fig 10.7 Admin Dashboard

The dashboard displays key metrics such as Total Users, Active Users, Inactive Users, and Total Service Requests. It also shows the status-wise breakdown of requests including Pending, In Progress, Resolved, and Cancelled, enabling administrators to track system performance in real time. The Quick Actions section allows administrators to manage users, configure service catalogs and categories, view all service requests, generate reports, and modify system settings. Navigation

options such as Users, Service Catalog, Categories, All Requests, Reports, Departments, Settings, and Audit Logs provide complete administrative control.

This module improves operational efficiency, ensures transparency, and enables effective monitoring and decision-making within the organization.



User Management

ADD NEW USER

Username	Name	Email	Department	Role	Status	Actions
durga	Durga Prasad	ddp9398@gmail.com	IT	Unknown	Active	
kasii	kasi p	kasi764096@gmail.com	Team	Unknown	Active	
sai	Sai Ram	kandregulasairam@gmail.com	HR Department	Unknown	Active	
admin	System Administrator	admin@servicedesk.com	IT	Unknown	Active	
phani	phani samavedam	phaniprince849@gmail.com	-	Unknown	Active	
Satish	Satish Bokka	satishbokka2002@gmail.com	CSE	Unknown	Active	
abhishekmadhura	Abhi` Madhura	abhishekmadhura39@gmail.com	-	Unknown	Active	
YVD Sai	Venkata Durga Sai Yarramsetti	yvdsai@gmail.com	IT	Unknown	Active	
Revathi	Revathi Jonnada	revathibunga511@gmail.com	CSE	Unknown	Active	

Fig 10.8 User Management

The User Management module of the Service Request System, where administrators can view all registered users. It displays user details such as username, email, department, role, and account status. Admins can add new users, edit existing user details, activate or deactivate accounts, and remove users if required. This module helps in maintaining proper user access and system security.

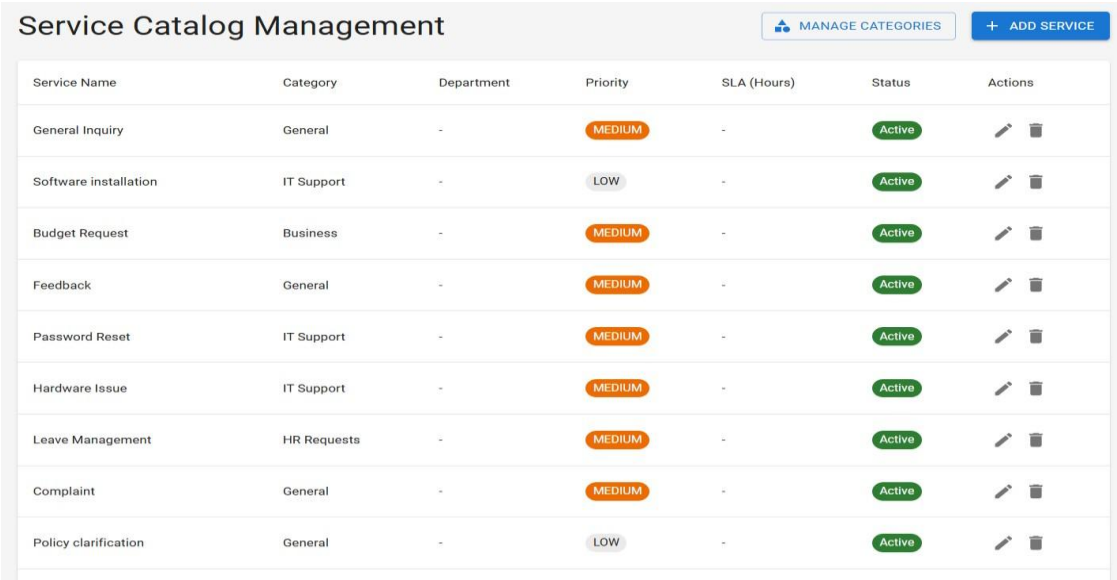


Fig 10.9 Service Catalog Management

Service Catalog Management module, where administrators manage all available services in the system. It displays service details such as service name, category, priority level, SLA hours, and current status. Admins can add new services, edit existing ones, or remove services when required. This module helps organize services and ensures proper request routing.

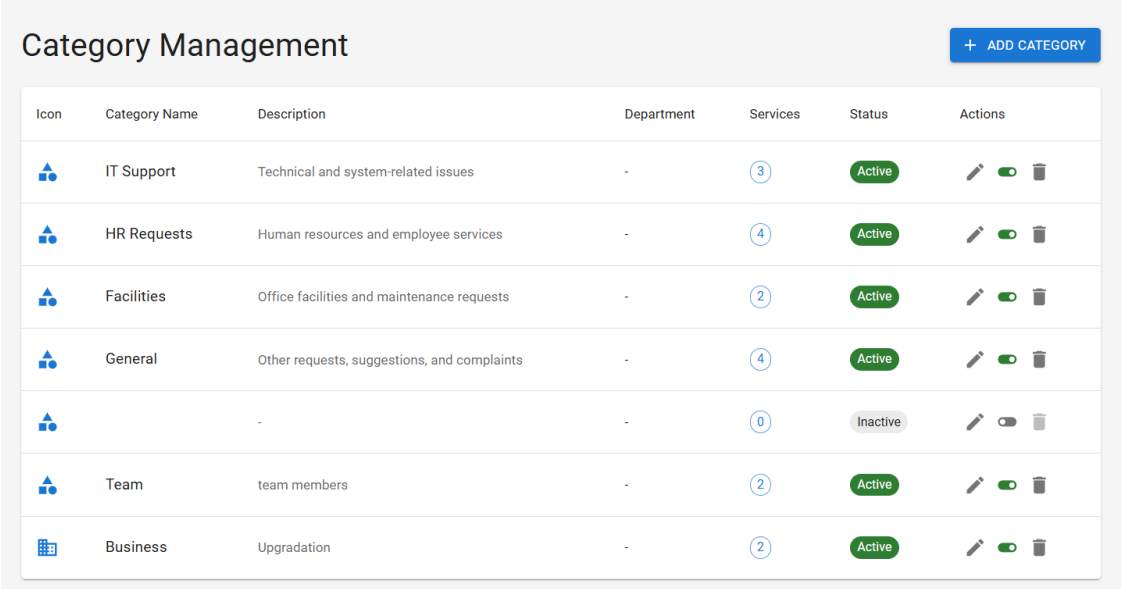


Fig 10.10 Category Management

Category Management module, where administrators manage different service categories in the system. It displays category name, description, associated services, department, and current status. Admins can add new categories, edit existing ones, activate or deactivate categories, and delete unused categories. This module helps organize services and improves request classification.

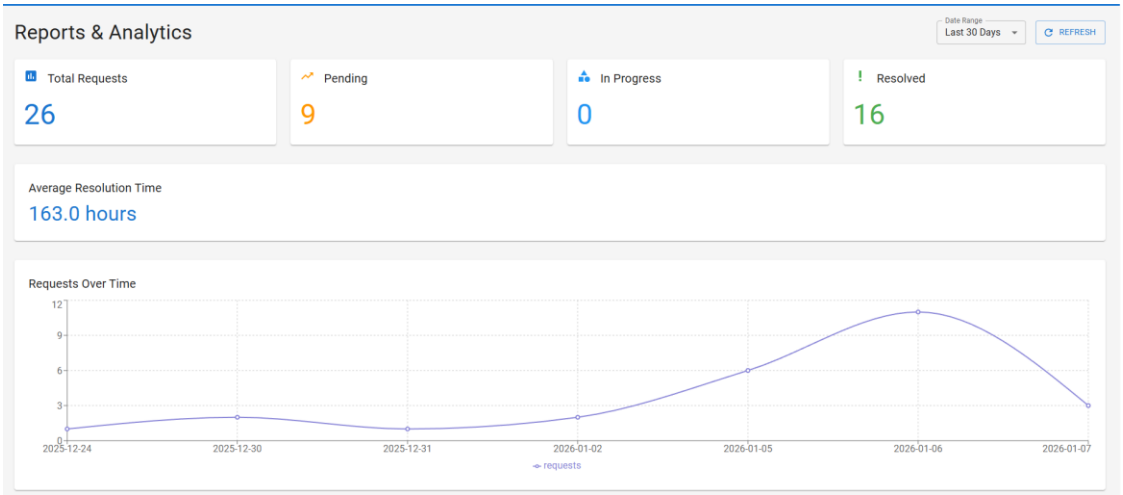


Fig 10.11 Reports & Analytics

Reports and Analytics module of the system, which provides an overview of service request performance. It shows total requests, pending, in-progress, and resolved requests along with the average resolution time. The graphical view helps administrators analyze request trends over time. This module supports better decision-making and performance monitoring

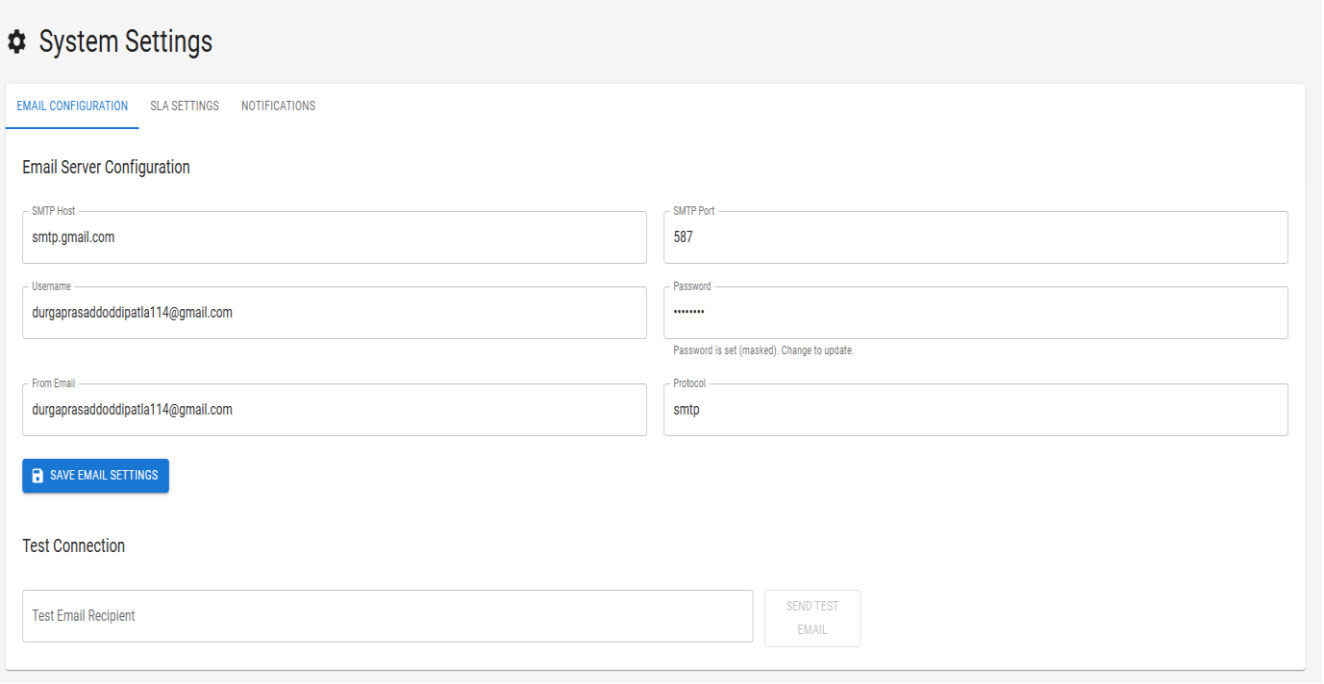


Fig 10.12 System Settings – Email Configuration

Allows the administrator to configure the email server used by the system for sending notifications. SMTP details such as host address, port number, login credentials, and protocol are provided to establish a secure email connection. Once configured, the system automatically sends emails for events like request creation, assignment, status updates, and resolution. The Test Connection feature helps verify whether the email settings are correct before saving. This module ensures reliable communication between users and administrators and improves system responsiveness.

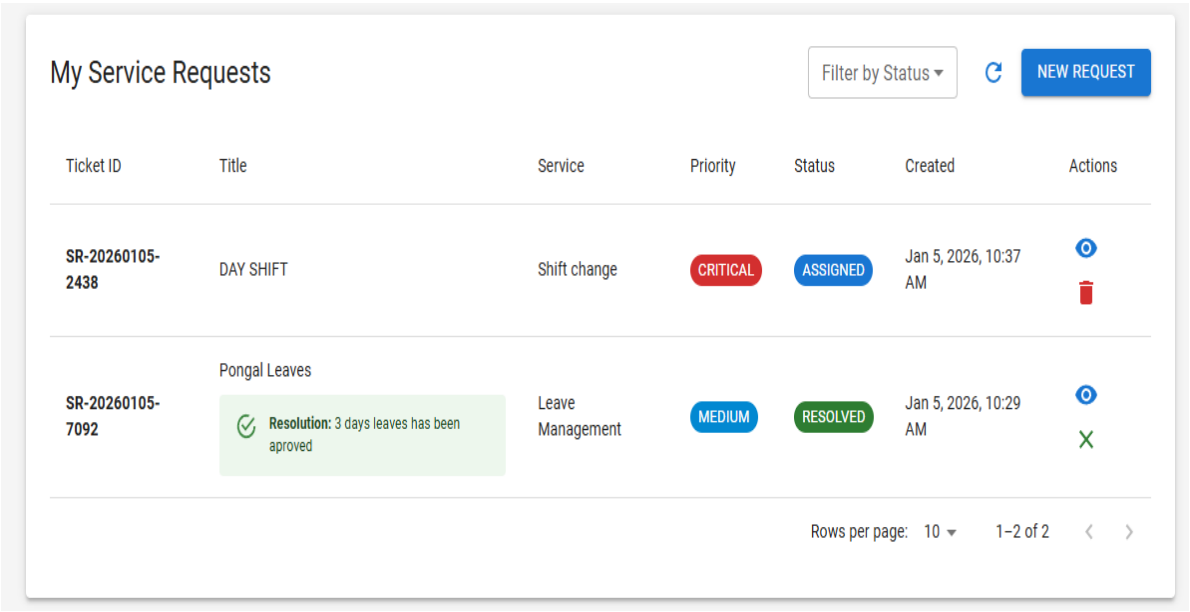


Fig 10.13 My Service Requests – Detailed View

shows the list of service requests submitted by the logged-in user along with their current status. It displays important details such as ticket ID, request title, service type, priority level, and creation date. Users can see whether a request is assigned, resolved, or still in progress. The resolution message provides clear confirmation once a request is successfully completed, improving transparency and user satisfaction.

CHAPTER – 11

CONCLUSION

The Service Request & Workflow Automation System has been successfully designed and implemented to provide an efficient and structured approach for managing service requests within an organization. The system replaces traditional manual processes with a centralized digital platform, allowing users to submit, track, and manage service requests easily. By automating workflows, the system reduces response time, improves accuracy, and enhances overall operational efficiency.

The application provides role-based access for users and administrators, ensuring security and controlled system usage. Users can register, log in, raise service requests, monitor request status, and receive timely notifications. Administrators can manage users, service categories, service catalogs, requests, and system settings through an intuitive dashboard. Features such as priority management, request assignment, status tracking, and reporting help administrators handle requests effectively and make informed decisions.

The integration of email notifications ensures continuous communication between users and administrators, keeping all stakeholders informed about request updates. The reports and analytics module provides valuable insights into request trends, resolution time, and system performance, which can be used to improve service quality and resource planning.

Overall, the Service Request & Workflow Automation System offers a scalable, secure, and user-friendly solution for service management. The project demonstrates the effective use of modern technologies such as Spring Boot, REST APIs, and database management to build a real-world enterprise application. This system can be further enhanced by adding features such as mobile access, AI-based request prioritization, and advanced analytics, making it suitable for deployment in real organizational environments.

CHAPTER – 12

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6. Spring Data JPA Documentation
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    Source code repository of the implemented project.
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    Provides reference implementations for enterprise-level Java projects

SERVICE REQUEST AND WORKFLOW AUTOMATION SYSTEM

With Student–Admin Integrated Management Platform

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[1]Student, [2]Student, [3]Student, [4]Student, [5]Student, [6]Assistant Professor
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[1]BVCEC, Odalarevu, India

***Abstract*:** In modern organizations, managing service requests manually through emails, spreadsheets, or paper-based forms often leads to delays, lack of transparency, and inefficient tracking. To overcome these challenges, this project presents a Service Request and Workflow Automation System developed using Java Full Stack technologies.

The system provides a centralized web-based platform where users can submit service requests digitally. Each request is automatically routed through predefined workflow stages such as validation, approval, task assignment, processing, and closure. Role-based access control ensures that users, administrators, and service staff have clearly defined permissions.

The Admin module allows user management, workflow configuration, task assignment, monitoring of request progress, and generation of reports. The User module enables request submission, status tracking, and real-time notifications.

The proposed system improves efficiency, reduces manual intervention, enhances transparency, and ensures structured workflow automation. The system is scalable, secure, and suitable for educational institutions, corporate offices, and enterprise environments.

***IndexTerms* – Service Request Management, Workflow Automation, Role-Based Access, Java Spring Boot, Web Application**

1. INTRODUCTION

In the digital era, organizations rely heavily on information systems to manage operational activities efficiently. One of the most common activities across organizations is handling service requests such as IT support, maintenance issues, approvals, and administrative services.
Traditional service request handling methods involve:

- Emails
- Phone calls
- Paper forms
- Manual tracking

These methods lack structured workflows and transparency, often leading to delays, lost requests, and miscommunication.
The proposed **Service Request and Workflow Automation System** provides a centralized digital platform where service requests are:

- Submitted online
- Assigned unique request IDs
- Routed automatically through predefined workflow stages
- Monitored in real-time

The system ensures automation, accountability, and efficient service management.

2. NEED FOR THE STUDY

The need for a Service Request and Workflow Automation System arises due to the following limitations in existing systems:

- Manual processing increases workload and human errors
- Lack of centralized tracking
- No predefined workflow stages
- Delayed approvals and task assignments
- Poor communication between departments
- Absence of real-time monitoring

3. LITERATURE SURVEY

Several systems have been developed for ticket management and service request tracking. Traditional systems include:

- Manual request registers
- Email-based ticketing
- Basic computerized ticket tools

Although these systems reduced paperwork, they lacked:

- Workflow automation
- Real-time notifications
- Role-based access control
- Analytics dashboards

Recent studies highlight the importance of workflow automation in improving efficiency and accountability. Modern frameworks like Spring Boot enable scalable and secure backend systems for enterprise-grade applications.
This project addresses the gap by implementing a complete automated workflow system using Java Full Stack technologies.

4. PROPOSED SYSTEM

The proposed system is a web-based application that automates the complete lifecycle of service requests.

Key Features:

- Online request submission
- Unique request ID generation
- Automated workflow routing
- Role-based authentication
- Task assignment and tracking
- Real-time notifications
- Reporting and analytics

Roles:

1. **User**
    - Submit service requests
    - Track request status
    - Receive notifications
2. **Admin**
    - Manage users
    - Configure workflows
    - Assign tasks
    - Monitor request progress
    - Generate reports

The system ensures structured and consistent processing of requests.

5. ARCHITECTURE OVERVIEW

The system follows a **three-tier architecture**:

1. Presentation Layer

- Developed using HTML, CSS, JavaScript, React
- Provides user interface for request submission and dashboards

2. Application Layer

- Java & Spring Boot
- Handles authentication, workflow logic, task routing

3. Database Layer

- MySQL
- Stores users, requests, workflow stages, logs

REST APIs enable communication between frontend and backend.

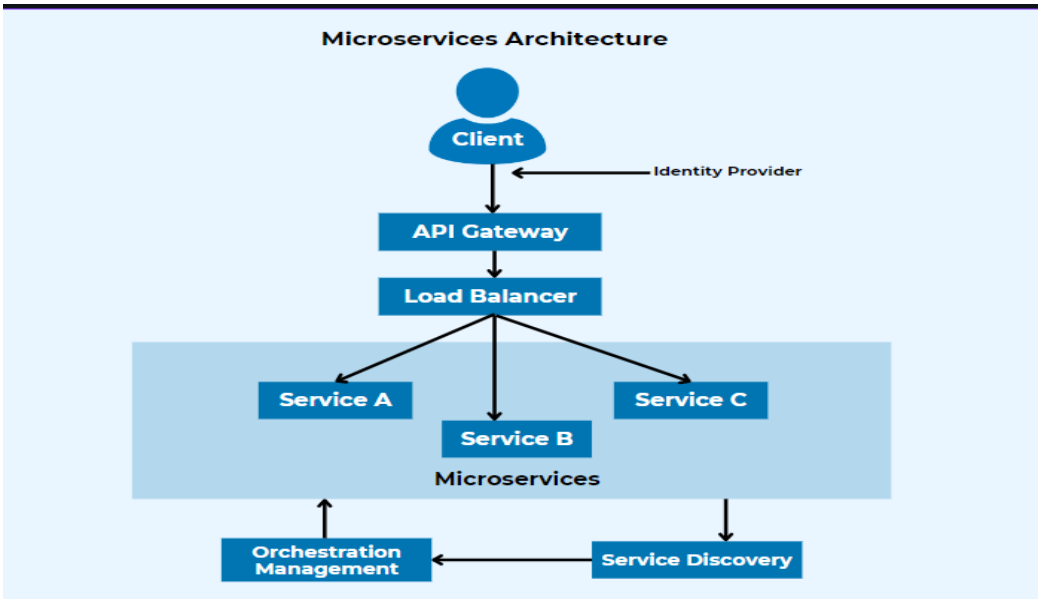

Fig 5. system architecture

6. MODULES EXPLANATION

6.1 Admin Module

- User Management
- Service Category Management
- Workflow Configuration
- Task Assignment
- Request Approval/Rejection
- Monitoring Dashboard
- Reports & Analytics

6.2 User Module

- Registration & Login
- Create Service Request
- View Request Status
- Track Workflow Progress
- Receive Notifications

6.3 Workflow Automation Module

- Automated routing
- Stage transitions
- Approval validation
- Escalation rules

6.4 Notification Module

- Email/Alert notifications
- Status updates
- Approval notifications

7. IMPLEMENTATION

The system was developed using:

- **Frontend:** HTML, CSS, JavaScript, React
- **Backend:** Java, Spring Boot, REST APIs
- **Database:** MySQL
- **Authentication:** Role-Based Login System
- **Server:** Embedded Apache Tomcat
- **Tools:** IntelliJ IDEA, VS Code, Postman

Development included:

- Secure authentication setup
- Workflow rule configuration
- CRUD operations for requests
- Dashboard analytics integration
- Testing and debugging

8. RESULTS & ADVANTAGES

8.1 Results

- Automated workflow execution
- Reduced manual intervention
- Real-time tracking of service requests
- Secure role-based access
- Improved communication
- Accurate data storage

The system successfully processes requests from submission to closure with transparency and efficiency.

8.2 Advantages

- Reduced paperwork
- Improved accountability
- Faster request processing
- Real-time notifications
- Secure data management
- Scalable architecture
- Suitable for enterprise deployment

9. CONCLUSION

The Service Request and Workflow Automation System provides a structured, automated, and centralized solution for managing service requests efficiently. By incorporating workflow automation, role-based access control, and real-time tracking the system improves operational efficiency and user satisfaction

The project demonstrates the effective use of Java Full Stack Technologies in building scalable enterprise solutions

10. FUTURE SCOPE

The system can be enhanced with:

- Mobile application integration
- Email & SMS notifications
- AI-based request prioritization
- SLA (Service Level Agreement) monitoring
- Cloud deployment
- Advanced analytics dashboard
- Integration with ERP systems

These improvements will make the system more powerful and enterprise-ready.

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